The Concept Allocation Zone

Tracking How Concepts Form Across Transformer Depth

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Version 2 corrects and retracts claims made in v1 (arXiv:2605.24856). Readers holding v1 should consult the "Changes from Version 1" section at the end of this paper for the full list of superseded values.

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Abstract

Concept formation in transformer language models is a depth-extended process, not a single-layer event: a concept becomes separable across one or more contiguous regions of the residual stream — its **Concept Allocation Zone** (CAZ). A CAZ is not a concept but the depth segment where the model organizes its geometry to make one separable — concepts may share a CAZ, and typically span multiple across depth; the companion GEM paper [Henry, 2026b] shows the separating direction continues to rotate within a CAZ before stabilizing past its boundary. We formalize the CAZ through three layer-wise metrics — Separation, Concept Coherence, and Concept Velocity — with automated boundary detection that applies no significance threshold to CAZ membership (every segment is a CAZ; "strong" vs. "gentle" is score, never a binary cut). Empirical validation across 35 models, 8 architectural families, and 7 concepts shows the separation curve $S(l)$ is frequently **multimodal**, and scored detection surfaces a further category of subtle allocation regions ("gentle CAZes") invisible to standard peak detection. The framework generates seven testable predictions (§5); its contribution is the instrument and the phenomena it surfaces — the scored detector, the three metrics, and the multimodal/gentle-CAZ findings — not the predictions themselves (full scorecard, §8). Released as the open-source rosetta_tools library (v1.3.1) [Henry, 2026, software].

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1. Introduction

The dominant paradigm in mechanistic interpretability extracts concept representations by identifying the single "best layer" — the residual stream depth at which a linear probe or difference-of-means (DoM) vector achieves maximum class separation [Zou et al., 2023; Arditi et al., 2024]. This heuristic is computationally convenient and empirically grounded. It is also, by design, a snapshot: it identifies the peak of a process rather than characterizing the process itself.

Transformers are iterative dynamical systems. Each layer applies a sequence of attention and MLP operations that *write* new information into the residual stream, modifying and extending what prior layers contributed [Elhage et al., 2021]. A concept observed at Layer 15 was shaped by Layers 10 through 14 before it; the best-layer heuristic tells us where the concept is most legible, not how it arrived there.

This paper introduces the **Concept Allocation Zone** (CAZ) framework, which extends the interpretability toolkit from anatomy — *where is the concept most visible?* — to dynamical flow — *how does the concept form?* The CAZ is defined as a region of model depth allocated to the

geometric expression of a concept — the interval within which that concept becomes measurably separable. A CAZ is not the concept itself — each CAZ is a depth-extended segment of the residual stream in which the model's geometry is organized to serve a concept.

The framework has immediate practical implications:

1. **Richer extraction.** CAZ/GEM (Geometric Evolution Maps; §4.2) diagnostics enable selecting a better single extraction layer (the handoff layer) than naive peak-layer probing (§4.4) — the value is in *which* layer CAZ/GEM identifies. Whether tracking depth this way outperforms a single-layer probe is a separate question the companion GEM paper addresses on its own terms [Henry, 2026b].
2. **Intervention depth is encoding-strategy-dependent.** Ablation at different points in the CAZ chain produces qualitatively different effects in some models, but in models with redundant encoding, suppression is layer-invariant (§5.1, §5.8) — so the framework characterizes *when* layer choice matters rather than prescribing a single optimal intervention depth. The original prediction that mid-CAZ is optimal (P1) is not supported (§5.1).
3. **Dark matter connection.** The *nonlinear* component of sparse-autoencoder (SAE) reconstruction error — the part Engels et al. [2025a] find is not linearly predictable from the input activation, a minority of the error norm — may partially correspond to in-progress concept construction within CAZes: transitional representations that resist linear decomposition at any single layer. This is a testable hypothesis; direct cross-validation of CAZ regions against SAE residuals has not been conducted in this work.
4. **Cross-model transfer.** Concept directions may be depth-matchable across architectures — a potential geometric basis for transferable interpretability tooling. Whether a single Procrustes rotation suffices to align all concepts simultaneously, and whether cross-architecture alignment is genuinely depth-matched, is the subject of dedicated follow-up work outside this paper's scope.
5. **Understanding alignment training.** CAZ profiles provide a lens for studying what alignment training changes in a base model's concept geometry — not whether concepts exist, but where and how they are allocated. In principle CAZ profiles support a metric for quantifying alignment-training distortion — the change in concept-selective feature count and alignment strength from base to instruct variant — which is left to follow-up work.
6. **Concept inventory.** By tracking which geometric directions are allocated at each CAZ and which remain unaligned with any human concept probe, the framework provides a systematic approach to cataloging what a model computes — both the named and the unnamed.

The paper's positive contribution is first-party: the scored detector and its three metrics, and the multimodal and gentle-CAZ phenomena they surface (§4–§6). The framework also generates seven specific, falsifiable predictions, fixed before this analysis began; §5 reports on all seven, giving the per-prediction verdicts and the model subset each was evaluated on, summarized in §8. Where they missed, following up on why produced some of the framework's most useful results — the two-regime encoding structure (§5.8), the gentle-CAZ category, and the handoff-layer question that grew into the companion GEM paper [Henry, 2026b]. The reference implementation is provided as rosetta_tools [Henry, 2026, software], an open-source Python library. We are explicit about the assumptions the framework inherits from the broader interpretability literature.

Corpus and analysis coverage. Two companion papers are cited throughout: the **GEM paper** [Henry, 2026b] (handoff-layer extraction) and the **validation paper** [Henry, 2026c] (large-scale causal ablation). Each uses a model set scoped to its analysis; the table below records the scope rationale for each:

Analysis	Coverage	Scope rationale
CAZ detection and predictions (this paper)	35 models, 7 concepts	29 base + 6 instruct; full architectural range, retained from the initial release
Direction-specificity ablation (this paper)	16 of 35 models	Extended to all 28 base models in [Henry, 2026c]
GEM handoff validation [Henry, 2026b]	29 base models, 17 concepts	Base models only; concept set expanded for handoff analysis
Full ablation sweep [Henry, 2026c]	28 base models, 17 concepts	Instruct variants excluded; systematic per-prediction evaluation

Individual predictions in §5 are evaluated on subsets of the 35-model corpus, and the subset is stated where each is reported: the P2 ordering recompute uses the **29 base models** carrying all seven concepts (§5.2); the P7 scale correlation uses **26 models** (§5.7); the chain-depth decay analysis uses **28 base models** for the remaining-depth result and **26 models** for the unembedding-clustering sub-result (§5.8, the latter's 182 measurements being 26×7); and the P3 width test uses **22 models** across 6 concepts (§5.8, n = 132). Each subset is the set for which the relevant artifact exists, not a selection on outcome.

The 7-concept corpus is this paper's evaluative paradigm. The seven concepts (credibility, certainty, causation, temporal_order, sentiment, negation, moral_valence) are the set on which the framework's predictions were pre-registered, and every result this paper computes *itself* — CAZ detection, the prediction scorecard (§5), the worked examples (§6), and both figures — is reported at that C=7 scope. The validation paper [Henry, 2026c] re-evaluates all seven predictions on the expanded 17-concept corpus (adding ten later behavioral and safety concepts), and the GEM handoff results [Henry, 2026b] use the same 17-concept set; those C=17 figures are *cited* here as the broader test and, where they differ, supersede this paper's C=7 assessment (§8).

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2. Relationship to Existing Work

The CAZ framework operates in the same space as several established interpretability methods. Each captures different aspects of model internals; the CAZ contribution is the layer-indexed dynamical view — it tracks *when* representations form, rather than only characterizing *what* they are.

2.1 Methodological Context

Logit Lens and Tuned Lens. The most direct methodological antecedents of the CAZ framework are the logit lens [nostalgebraist, 2020] and the tuned lens [Belrose et al., 2023]. Both project intermediate residual-stream activations into a fixed interpretive space — the unembedding matrix for logit lens, a per-layer trained affine probe for tuned lens — to produce a layer-by-layer trajectory of the model's evolving prediction. CAZ shares the core commitment to tracking representations across depth rather than characterizing a single best layer, and any claim about layer-indexed concept dynamics necessarily inherits from this line of work. The distinction is what is being projected and asked. Logit and tuned lens project onto output-vocabulary space and ask *what token would this layer predict if the model stopped here?*; the CAZ framework projects onto concept-contrast

directions obtained from external stimulus pairs, then tracks the resulting class separation across depth to ask *where in the stack is this concept being allocated?* Concepts without a canonical lexicalization — credibility, refusal, moral valence — do not project cleanly onto vocabulary axes and fall outside the natural scope of logit-lens interpretation, which motivates the contrast-direction formulation used here. Calibrating the velocity metric against Tuned Lens in settings where concept and vocabulary are tightly coupled is left to future work.

Sparse Autoencoders (SAEs) decompose activations at a fixed layer into interpretable monosemantic features [Cunningham et al., 2023; Bricken et al., 2023]. SAEs answer "what features exist at layer L?" The CAZ framework answers "how does the feature at layer L relate to the feature at layers L−1 and L+1?" The approaches are complementary: SAE features at a given layer are a snapshot; CAZ tracking reveals which snapshots are part of the same evolving computation. Engels et al. [2025a] decompose SAE "dark matter" — the unexplained variance in activations — and find that about half of the error vector, and more than 90% of its norm, is *linearly predictable* from the input activation; the residual that resists linear prediction is a distinct and much smaller "nonlinear error" component. It is that nonlinear component, not dark matter as a whole, that the CAZ framework offers a candidate mechanism for: in-progress concept construction within a CAZ produces transitional representations that are neither the input feature nor the output feature. These transitional states may be precisely what resists decomposition at any single layer.

Centered Kernel Alignment (CKA) [Kornblith et al., 2019] measures representational similarity between layers or models by comparing activation kernel matrices. CKA provides a global similarity score between two representation spaces but does not identify *which* features are shared or how they evolve across depth. The CAZ framework tracks individual concept directions layer by layer, producing a trajectory rather than a scalar comparison. Whether CKA serves as a complementary validation tool — high within a CAZ where the representation is being refined, low at inter-CAZ saddle points where it is being reallocated — is left to future work.

Linear probing [Belinkov, 2022; Alain & Bengio, 2017] trains classifiers on activations at each layer to measure concept presence. Probing accuracy curves are closely related to the separation metric $S(l)$ — both measure how distinguishable two classes are at a given depth. The CAZ framework adds the velocity metric (rate of change of separation) and the coherence metric (geometric quality of the separating direction) — formalized in §4 — which together identify where a concept is present *and* where it is actively being constructed. Probing also requires training a classifier per layer; the CAZ metrics are computed directly from activation statistics. Prior work has also established that this depth-ordered probing view recovers a syntactic-to-semantic gradient — shallow layers encode surface/syntactic features, deeper layers semantic ones [Tenney et al., 2019] — which is the empirical basis for our Prediction 2 (relative cross-architecture ordering, §5.2).

Concept activation vectors (TCAV). Concept directions in activation space trace to concept-based explanation methods such as TCAV [Kim et al., 2018], which derives a concept vector from a linear classifier over concept-present vs. concept-absent activations and measures a concept's influence on predictions via directional derivatives. Our concept directions are instead obtained by difference-of-means over contrastive pairs (cf. [Zou et al., 2023]; [Marks & Tegmark, 2024]), and our object of study is the *depth* at which a concept is maximally separable rather than its sensitivity to a downstream prediction — so TCAV is an adjacent concept-based-explanation precursor, not the construction the CAZ directions derive from.

Representation Engineering (RepE) [Zou et al., 2023] extracts concept directions for honesty, morality, power-seeking, and related concepts via contrastive stimuli, then uses those directions

for monitoring and steering. The CAZ framework bears on RepE's operational decisions, with the caveat that everything below comes from the validation paper's geometric ablation protocol and has not been validated for behavioral steering ([Henry, 2026c] §6.8). First, the CAZ profile identifies where a concept is being actively constructed versus where it is established, offering a candidate basis for choosing intervention depth. Second, the model's encoding strategy determines whether layer selection matters at all: in models with redundant encoding, the concept direction persists across all post-CAZ layers, suggesting layer choice matters less there; in models with sparse encoding, the CAZ peak concentrates the measurable signal, and intervening elsewhere may miss the target. Third, the scored CAZ detector (§4) reveals that a concept may have multiple candidate layers — whether RepE steering applied at a gentle CAZ (§4) produces different behavioral effects than steering at the dominant peak is an open question.

Activation patching and causal scrubbing. Activation patching — replacing activations at a specified layer and component with those from a counterfactual run [Vig et al., 2020; Meng et al., 2022; Wang et al., 2023] — is the canonical toolkit for testing whether a proposed locus genuinely carries a computation. Causal scrubbing [Chan et al., 2022] formalizes this as a discipline for rigorously testing mechanistic hypotheses; path patching [Goldowsky-Dill et al., 2023] extends the method to circuit-level attribution. Any claim of the form "the concept is allocated at layer l" is a candidate for validation under this family of methods, and the ablation protocol used in the validation paper [Henry, 2026c] is a direct descendant. The CAZ contribution is upstream of patching rather than parallel to it: the velocity-based boundary and score-based region detectors (§4) produce a principled, concept-specific choice of *which* layers to patch, replacing exhaustive layer sweeps with a targeted hypothesis about the allocation zone. The two methods are therefore complementary — CAZ identifies candidate layers; activation patching tests whether the identified layers are causally necessary.

2.2 Related Empirical Findings

MLP feed-forward layers as key-value memories. Geva et al. [2021, 2022] showed that FFN layers function as key-value memories that incrementally promote concepts into the vocabulary space across depth. The CAZ metrics ($S(l)$, $v(l)$, $C(l)$) are defined on the residual stream and are agnostic to whether promotion happens through attention or feed-forward paths; connecting CAZ boundaries to specific FFN write operations is a testable mechanistic question not pursued here.

Manifold interpretability. Gurnee et al. [2026] found curved manifolds in early layers for character counting and explicitly called for unsupervised geometric discovery methods. The CAZ framework provides a formalism for identifying *where* in the layer stack such manifolds crystallize — the allocation zone is precisely where curved manifold structure should be most geometrically coherent.

The geometry of refusal. Arditi et al. [2024] and Wollschläger et al. [2025] establish the geometric structure of refusal — a single removable direction (abliteration) and multi-dimensional concept cones, respectively. The CAZ framework extends this by asking *when* the geometry forms, alongside *what* it is, and whether that temporal structure bears on the choice of intervention point (§5.1).

Multi-dimensional concept structure. Engels et al. [2025b] found circular multi-dimensional representations for temporal concepts that are not decomposable into independent one-dimensional SAE features. Wollschläger et al. [2025] showed refusal occupies polyhedral concept cones. These findings establish that rich geometric structure exists; the CAZ framework provides a layer-indexed account of when such structures crystallize.

Concept geometry in production models. Sofroniew et al. [2026] demonstrate that Claude Sonnet 4.5 develops internal representations of 171 emotion concepts whose geometry is stable across model depth (particularly early-middle to late layers) and generalizes across contexts and behaviors. The representations causally influence model behavior — including downstream effects on reward hacking, sycophancy, and blackmail — confirmed via activation steering. Concept vectors are extracted by averaging residual stream activations across stories per emotion and subtracting the cross-emotion mean; the resulting representations cluster by semantic similarity with valence and arousal as primary axes, mirroring human psychological structure. This result provides direct support for two foundational claims underlying the CAZ framework: that transformers develop stable, generalizable geometric representations of human-interpretable concepts in their residual streams — representations that hold across diverse inputs rather than being probe artifacts — and that this structure is causally meaningful rather than a mere correlate of surface features (this paper's own causal evidence for CAZ specifically is discussed in §7).

The Platonic Representation Hypothesis. Huh et al. [2024] proposed that models trained on different data and architectures converge on shared representations. The CAZ framework enables a depth-stratified test of this hypothesis: rather than measuring global alignment, we can ask whether convergence differs at shallow versus deep processing stages.

Universal neurons. Gurnee et al. [2024] identified individual neurons in GPT-2 that activate on consistent input features across five independent training runs — direct evidence that specific representational structure is stable under retraining. This is a neuron-level instantiation of the architectural-convergence phenomenon the Platonic Representation Hypothesis [Huh et al., 2024] addresses at the level of whole representation spaces. The CAZ framework predicts a depth-stratified extension of both claims: if concepts allocate at architecturally stable proportional depths, then the layers hosting universal neurons for a given concept should correspond to that concept's CAZ peak in proportional terms, and cross-architecture alignment of concept directions should be strongest when evaluated at matched CAZ depths rather than at matched absolute layer indices. Universal-neuron discovery and CAZ-profile matching would then be two lenses on the same underlying regularity at different granularities.

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3. Background

3.1 The Residual Stream and Concept Representation

The residual stream formulation [Elhage et al., 2021] treats each layer's output as an additive contribution to a shared communication channel. Attention heads and MLPs read from and write to this stream; the final residual vector is projected onto the unembedding matrix to produce logits. This architecture makes layer-by-layer tracking of concept geometry natural: we can ask, at each layer l, how well the current residual stream separates two contrastive classes.

3.2 Difference-of-Means and Linear Artificial Tomography

DoM extracts a concept direction $V_{\text{concept}} \in \mathbb{R}^d$ as the normalized difference between class-conditional mean activations at the chosen layer [Zou et al., 2023]. Linear Artificial Tomography (LAT) uses a similar contrastive approach. Both methods produce a single vector at a single depth — a precise and useful representation of where the concept is most geometrically legible. The CAZ framework asks what additional information about concept formation might be recoverable from the layers

surrounding that peak.

3.3 Abliteration and Intervention Depth

Arditi et al. [2024] demonstrated that refusal behavior across 13 open-source models is mediated by a single direction removable via weight orthogonalization ("abliteration"), and report that this intervention disables refusal with minimal effect on other capabilities. The CAZ framework motivates a systematic study of intervention depth: rather than selecting a single layer by hyperparameter search, the CAZ profile identifies where concepts are being actively constructed versus where they are established, offering a candidate basis for choosing intervention points (§5.1 reports the pre-registered mid-CAZ version of this idea as not supported).

3.4 The Emerging Geometric Program

Gurnee et al. [2026], Engels et al. [2025b], and Wollschläger et al. [2025] — surveyed in §2.2 — establish a growing body of evidence for rich geometric structure in activation space, and their authors have explicitly called for unsupervised methods to detect and characterize it. The CAZ framework complements this program by providing a layer-indexed account of *when* such structures crystallize, in addition to *where*. The companion GEM paper [Henry, 2026b] extends this same question to extraction itself, tracking how a concept's separating direction rotates within a CAZ before it settles (§4.4).

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4. The Concept Allocation Zone

A CAZ is a **locator**: a coordinate on model depth that brackets where a concept is assembled — not the concept itself, and not a discrete functional module.

Operationally, a CAZ is the segment of the residual stream between two saddle points of the separation curve $S(l)$, around a local separation peak (§4.3). The detector partitions depth into such segments and scores each; the score grades **geometric salience** (Major→Gentle; see Terminology) — not causal importance (the validation paper [Henry, 2026c] §6.4), and not membership: every segment is a CAZ, and "strong" versus "gentle" refers to score. Because the segmentation covers the full depth, a CAZ is depth-*extended*, not depth-*localized* — in the limit a single segment can span the entire network (the GPT-2-XL causation case: one region across all 48 layers, §6.1) — so the score says how sharply the coordinate is drawn, not how much computation the network performs there.

Where the causal work occurs is a separate question the CAZ does not answer. It is displaced from the separation peak (the validation paper [Henry, 2026c] §6.4), and in redundant-encoding models causal efficacy extends past the CAZ boundary entirely — the optimal intervention point is post-CAZ, not mid-CAZ (§5.1, §5.8). The CAZ marks the zone; it does not claim to locate the computation within or beyond it.

The concept is the human label we project onto the geometry. A single CAZ may host multiple concepts simultaneously — a depth-indexed analog of polysemanticity, where a shared geometric resource serves more than one concept rather than one feature serving more than one meaning ([Henry, 2026c]) — and a single concept typically participates in multiple CAZes across depth (mean 2.20 per concept per model). When we say "credibility has a CAZ at layer 10," we mean layer 10

falls within a segment allocated to credibility's geometric expression — not that layer 10 "contains" credibility.

Terminology

Term	Definition
CAZ	A locator — a coordinate on model depth that brackets where a concept is assembled. Operationally, a scored saddle-to-saddle segment of the separation curve (§4.3); not the concept, and not a functional module
CAZ Profile	The full sequence of CAZes for one concept in one model
Major CAZ	CAZ with score > 0.5 — the highest composite-salience band; score reflects prominence, coherence, and width jointly (§4.3), not spatial narrowness — a Major CAZ can span an entire network
Strong CAZ	CAZ with $0.2 <$ score ≤ 0.5
Moderate CAZ	CAZ with $0.05 <$ score ≤ 0.2
Gentle CAZ	A low-salience CAZ (score < 0.05), invisible to standard peak detection; the validation paper finds these regions causally active under geometric ablation in most cases (§7) — a direct illustration that the score grades salience, not causal importance
Embedding CAZ	CAZ at the earliest stored (transformer-block) layers, driven by near-lexical/token-level features rather than deep computation. The pre-transformer embedding output is excluded from extraction, so this denotes the earliest *block* layers, not the raw embedding (see §4.1)
Active CAZ	CAZ within the transformer layers, driven by attention and MLP computation. Active CAZes are the primary subject of this framework
CAZ score	Composite metric: (prominence / mean separation) $\times$ coherence boost $\times\sqrt{\text{width}/N}$ (full form in §4.3)
Separation $S(l)$	Within-class-scale-normalized centroid distance between contrastive classes at layer l
Coherence $C(l)$	Explained variance ratio of the primary separating direction at layer l
Velocity $v(l)$	Smoothed rate of change of $S(l)$ across layers
Directional Stability $DS(l)$	Layer-to-layer cosine similarity of the dominant concept direction $\hat{d}(l)$; near 1.0 means the direction barely moved; a sharp drop marks a handoff event (§4.2)

4.1 Concept Lifecycle

By tracking the residual stream across model depth, concept formation is empirically distinguishable as a sequence of allocation regions rather than a single allocation followed by decay.

Early layers (Context and Syntax)

In early layers, the residual stream primarily resolves local context, grammar, and surface token relationships. Projecting contrastive datasets into this space generally produces heavily entangled activations; the separation metric is near zero — the contrastive classes are not yet linearly distinguishable at these depths.

However, the scored detector applied to this paper's own corpus does find CAZes at the earliest stored layers (0–1) in tens of concept × model pairs, concentrated in concepts with strong lexical cues. Credibility is the clearest case: using an independent method (surface lexical cues, not the scored detector), the validation paper separately resolves credibility below 25% depth in 9 of 28 models ([Henry, 2026c] §3.3). **Layer-indexing convention.** Extraction excludes the model's pre-transformer embedding output (HuggingFace `hidden_states[0]`): stored layer 0 is the *first transformer block's* output, stored layer l is the output of block $l{+}1$, and depth percentages are taken over the N block outputs. These earliest-layer CAZes therefore sit at the first one or two transformer blocks, not at the raw embedding. Most are gentle-to-moderate (score < 0.1) and plausibly reflect near-lexical structure already present in early layers — concept-associated tokens carrying distinctive early-block geometry, a property of the tokenizer and training corpus rather than deep computation; direct token-embedding alignment does not confirm this account (§5.6). We term these **embedding CAZes** — a label for their near-lexical, earliest-layer character, *not* a claim that they occur before any transformer processing (the literal embedding layer is not in the extracted tensor) — to distinguish them from **active CAZes** where the model's attention and MLP computations allocate geometry to serve a concept. Practitioners using CAZ profiles should be aware that embedding CAZes may not respond to the same interventions as active CAZes.

CAZ Chain (Concept Allocation)

As the residual stream deepens, geometric directions are allocated to concepts in contiguous depth regions — the CAZes. A persistent direction in activation space may serve one concept at shallow layers, be reallocated to a different concept at a mid-depth CAZ, and reallocated again at a deeper one. For a given concept, the separation metric $S(l)$ reflects when that concept holds a strong claim on one or more geometric directions; a single concept typically participates in multiple CAZes across depth (mean 2.20 per concept per model under scored detection), ranging from major CAZes (score > 0.5) to gentle allocation regions (score < 0.05; Terminology).

Late layers (Logit Projection)

In the final layers, the model transitions from abstract representation to concrete next-token prediction. The residual stream is projected toward the unembedding matrix. Concept geometry may degrade or re-entangle as abstract directions give way to vocabulary-specific structure. This is the only phase where separation genuinely decays; earlier apparent "decay" between CAZ peaks may instead reflect reallocation of the direction to a different concept.

4.2 Layer-Wise Metrics

Let $h_l^{(i)} \in \mathbb{R}^d$ be the residual stream activation at layer l for sample i, and let A, B be contrastive classes with conditional means $\bar{h}_A^{(l)}, \bar{h}_B^{(l)}$ and within-class covariance matrices $\Sigma_A^{(l)}, \Sigma_B^{(l)}$.

Separation Metric

We define the separation at layer l as a within-class-scale-normalized centroid distance — a practical separability proxy in the spirit of Fisher's discriminant analysis [Bishop, 2006, §4.1.4], though, as noted below, not Fisher's criterion itself:

$$S(l) = \frac{\left\|\bar{h}_A^{(l)} - \bar{h}_B^{(l)}\right\|_2}{\sqrt{\frac{1}{2}\left(\mathrm{tr}(\Sigma_A^{(l)}) + \mathrm{tr}(\Sigma_B^{(l)})\right)}}$$

In plain terms: separation asks "if I gave you a sentence and asked whether it expresses credibility or not, how easily could you tell from the model's internal state at this layer?" A high $S(l)$ means the model's activations for credible and non-credible text have moved far apart relative to how spread out each group is. A low $S(l)$ means the two groups are still jumbled together. Tracking $S(l)$ across layers reveals where the model begins to "make up its mind" about a concept.

Raw centroid distance is misleading when cluster dispersion varies across layers. Early layers tend toward diffuse, high-variance representations; normalization by within-class spread corrects for this. Mahalanobis distance [Mahalanobis, 1936] would account for full covariance structure but requires estimating and inverting per-class covariances, which is ill-conditioned in high-dimensional activation spaces without shrinkage regularization — a route we do not pursue here. Our scalar denominator normalizes only by the *total* (isotropic) within-class scale, $\frac{1}{2}(\mathrm{tr}\,\Sigma_A + \mathrm{tr}\,\Sigma_B)$, not by scatter along the discriminant direction. Consequently $S(l)$ is **not** Fisher's discriminant criterion, and is not guaranteed monotone with linear separability under general (anisotropic) covariance: two layers with equal $S(l)$ can differ in Bayes-optimal separability, and residual streams are known to be anisotropic (a few rogue dimensions dominate $\mathrm{tr}\,\Sigma$). We adopt it as a closed-form, training-free separability proxy that is numerically stable and comparable across layers, not as a Bayes-optimal measure.

Concept Coherence

Separation alone is insufficient: two classes could exhibit identical centroid separation while one forms a tight cluster and the other a diffuse cloud. We track Concept Coherence as the explained variance ratio of the first principal component of the full pooled activation matrix at each layer:

$$C(l) = \frac{\lambda_1^{(l)}}{\sum_i \lambda_i^{(l)}}$$

where $\lambda_i^{(l)}$ are the eigenvalues of the covariance of all activations — positive and negative class combined — at layer l. Concretely, $C(l)$ is computed by fitting PCA on the concatenated $[h_A; h_B]$ matrix and reading off the explained variance ratio of the first principal component. A high $C(l)$ means the dominant source of variation across both classes is a single direction — the representation is geometrically concentrated. A low $C(l)$ means variation is spread across many dimensions — the representation is diffuse. A concept is *well-formed* when both $S(l)$ and $C(l)$ are high: the classes are far apart *and* the separating direction is geometrically clean.

In plain terms: coherence asks "is the concept encoded as a single clean direction, or is it smeared across many dimensions?" **Precision note**: because $C(l)$ is fitted on the pooled (both-class combined) activation matrix, it measures the dominant direction of total variance, which may

correspond to within-class spread rather than between-class separation. $C(l)$ is most interpretable in combination with $S(l)$: high $C(l)$ with high $S(l)$ confirms the dominant pooled direction is also the class-separating direction; high $C(l)$ with low $S(l)$ indicates activation concentration on a direction that is not class-discriminative. Figure 2 shows a worked example of the two curves moving apart: coherence peaks well before the deepest separation peak and is already declining by the time $S(l)$ reaches it.

Concept Velocity

To characterize the rate of geometric divergence between layers, we compute a smoothed first difference of the separation curve. Because raw layer-to-layer differences are noisy, we first apply a centered moving-average of width w to $S(l)$ and then take the adjacent-layer difference:

$$v(l) = \tilde{S}(l) - \tilde{S}(l-1), \qquad \tilde{S}(l) = \frac{1}{w} \sum_{j=-\lfloor w/2 \rfloor}^{\lfloor w/2 \rfloor} S(l+j)$$

where $\tilde{S}$ is S box-smoothed over a centered window of width w. The centered sum as written has $2\lfloor w/2 \rfloor + 1$ terms, so it matches the $1/w$ normalization only for odd w — satisfied by the fixed default $w = 3$ used throughout this paper (reference implementation: `rosetta_tools.caz.compute_velocity`). At the endpoints the window is zero-padded (numpy `same`-mode convolution), which mildly attenuates $\tilde{S}$ within $\lfloor w/2 \rfloor$ layers of each boundary — affecting only the diagnostic velocity trace, not the CAZ boundaries, which are taken from $S(l)$ directly. All results in this paper use the library's fixed default $w = 3$; a depth-adaptive setting $w = \max(1, \lfloor N/24 \rfloor)$ is also available but was not used for the corpus reported here. Velocity is reported as a trajectory *diagnostic* — it shows where separation is rising or falling — but the CAZ boundaries used throughout this paper are taken from the peaks and saddle points of $S(l)$ directly (§4.3, Multi-Region Detection), not from a velocity threshold.

In plain terms: velocity asks "is the concept forming right now, or has it already formed?" Positive velocity means separation is increasing — the model is actively constructing the concept at this layer. Negative velocity means separation is decreasing — the concept is being degraded or reallocated. Zero velocity means nothing is changing; velocity turns positive as allocation begins and negative as it ends. In the unimodal single-region case (§4.3) this profile can bound a CAZ directly; the scored multi-region detector used for the results in this paper instead places boundaries at the saddle points of $S(l)$, with velocity retained as a visualization aid.

Directional Stability

The three metrics above track *how strongly* a concept is expressed. $DS(l)$ tracks *which direction* it occupies. Let $\hat{d}(l)$ be the normalized dominant direction (dom_vector) at layer l — the unit vector from negative to positive class centroid. Directional stability is:

$$DS(l) = \hat{d}(l) \cdot \hat{d}(l-1)$$

$DS(l)$ near 1.0 means the dominant direction barely moved between consecutive layers; the model is encoding the concept stably. A sharp drop marks a **handoff** — a layer at which the geometric basis of the concept shifts substantially. The Geometric Evolution Map (GEM) framework [Henry, 2026b] reads the *settled* direction at each CAZ segment's final layer and evaluates it at the handoff layer — the first layer past the segment's saddle boundary; the handoff layer is derived from that boundary

(§4.4), not detected from a drop in $DS(l)$. The GEM angular velocity $\omega(l) = 1 - |DS(l)|$ is the complement, using the absolute value to treat direction-sign ambiguity as zero rotation rather than full rotation. The two readouts therefore differ on a pure sign flip by design — a sharp drop in signed $DS(l)$, but zero rotation in $\omega(l)$ — because antipodal directions span the same one-dimensional axis.

$DS(l)$ is conceptually independent from $S(l)$: separation can be high while the direction is still rotating (during active CAZ assembly), and separation can be low while the direction is stable (between concept peaks). The two signals decompose model activity into *what* is being encoded and *how stably*, respectively.

4.3 CAZ Boundary Detection

Single-Region Detection (Velocity-Based) When the $S(l)$ curve is unimodal, CAZ boundaries are derived from the velocity profile:

- **CAZ Entry** (l_{start}): The first layer where $v(l)$ exceeds a sustained positive threshold $\theta_+ = 0.5 \times \max_l v(l)$, maintained for at least two consecutive layers. Note: θ_+ uses the global maximum of $v(l)$ across all layers of the model, including any negative-velocity portions.
- **CAZ Peak** (l_{max}): The layer where $S(l)$ reaches its absolute maximum. This corresponds to the "best layer" of conventional interpretability.
- **CAZ Exit** (l_{end}): The layer where $v(l)$ becomes consistently negative for at least two consecutive layers, marking the onset of post-CAZ degradation.

The conventional best-layer heuristic extracts V_{concept} at l_{max}. CAZ-aware extraction uses the full interval $[l_{\text{start}}, l_{\text{end}}]$.

On the temporal_order example in Figure 1 below, per-CAZ ablation confirms both regions are causally active rather than an artifact of the peak-vs-peak-distal comparison §7 flags as partly circular: suppression reduces separation by 46.5% of separation at the shallow CAZ and 77.5% at the deep one — a non-tautological check, since it compares each region's own ablated state to its own intact state rather than comparing selected-for-high-separation layers to selected-for-low-separation ones.

Multi-Region Detection (CAZ Profiles) Empirical analysis across 35 models reveals that the $S(l)$ curve is frequently **multimodal** — a single concept can produce multiple significant local maxima at different depths. In these cases, the velocity-based boundary detector wraps a single contiguous zone around the global maximum and is blind to secondary peaks.

The **CAZ Profile** generalizes the single-region CAZ to a sequence of allocation regions:

1. Detect all significant local maxima in $S(l)$ using prominence-based peak detection.
2. Identify **saddle points** — the local minima between consecutive peaks — as natural region boundaries.
3. Each region spans from one saddle to the next, with the first region starting at layer 0 and the last ending at the final layer.

This segmentation **tiles the full depth** — every layer belongs to exactly one region — so "CAZ" names a segment of the partition, not a sparse subset of privileged layers; a region's *strength* is carried by its composite score (below), not by membership. This totality is a property of *this detector as configured*, not a theorem about CAZes in general: the detector emits at least the global

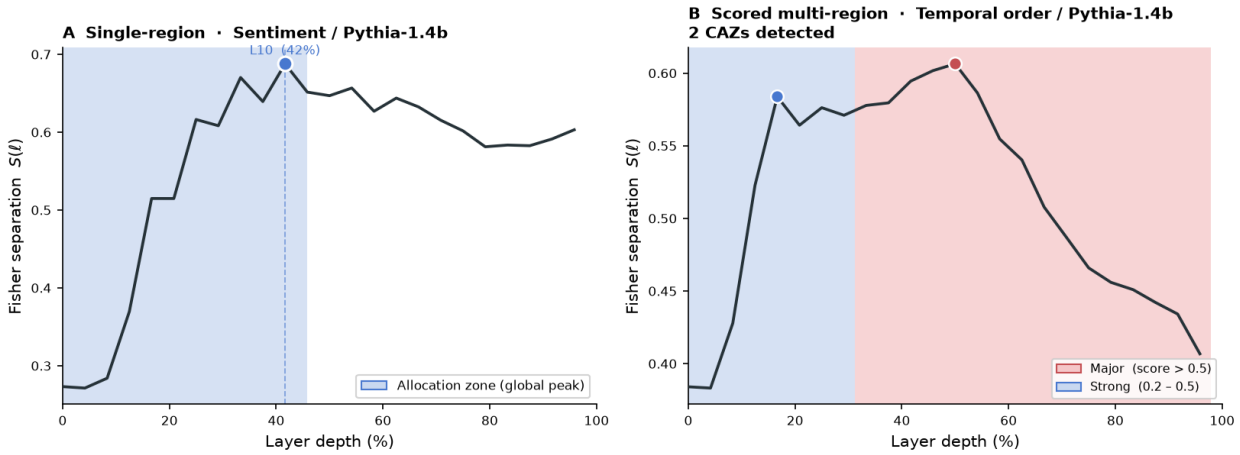

Figure 1: Single-region detection (left) versus scored CAZ profile (right). The single-region detector brackets one allocation zone around the global separation peak for sentiment in Pythia-1.4B (peak L10, 42% depth). The scored detector, applied to temporal_order in the same model, resolves two CAZes at geometrically distinct depths (§4.5): a shallow strong CAZ (L4, 17% depth, score 0.47) and a deep major CAZ (L12, 50% depth, score 0.65) — the single-region method would bracket only the deeper peak and miss the shallow one. N=250 contrastive pairs, $w = 3$ velocity smoothing (diagnostic only), final-token probing, scored detection defaults (0.5% prominence floor, 3% valley-merge threshold).

separation-peak region as a fallback when no prominence-qualifying peak is found and applies no score filter, so coverage is complete by construction (verified across the GEM paper's corpus, 493 concept × model pairs over 29 base models × 17 concepts; [Henry, 2026b]). The degenerate limit of this totality is therefore a *single* segment spanning the whole network (the GPT-2-XL causation case), never zero coverage; a differently-constructed segmenter need not tile. Note also that the first and last boundaries are imposed by the model's endpoints rather than detected from a saddle; in particular the final region always terminates at the last layer. Because the partition is total, there are no "non-CAZ layers" in an absolute sense: where this paper and its companions contrast CAZes against a peak-distal baseline (the peak-suppression enrichment; the gentle-CAZ causal checks), the comparator is defined *relative to peaks* — the validation paper uses layers more than three from any CAZ peak ([Henry, 2026c] Table 8). The size of that peak-distal comparator varies with model depth, a dependence the enrichment comparison must account for (see [Henry, 2026c]).

A CAZ Profile is characterized by: - **n_regions**: Number of distinct allocation regions (1 = unimodal, 2+ = multimodal) - **peak region**: The region with the highest peak separation - Per-region: start, peak, end, width, peak separation, coherence, rise/fall asymmetry

Scored Detection The prominence-based detector requires a threshold to determine which peaks are significant. A fixed threshold (e.g., 10% of global max separation) is arbitrary and risks discarding subtle but causally active allocation regions. We replace it with a composite scoring system:

$$\text{CAZ score} = \frac{\text{prominence}}{\bar{S}} \times \left(1 + \frac{C(l_{\text{peak}})}{\bar{C}}\right) \times \sqrt{\frac{\text{width}}{N}}$$

where **prominence** is the scipy peak-prominence value — the height of the peak above the higher of its two surrounding saddle points — and peaks are required to clear a minimum prominence of 0.5% of the global maximum $S(l)$ (the `prominence=` selection threshold passed to `scipy.find_peaks`: peaks below it are not detected, rather than clamped); $\bar{S}$ is the global mean separation across all layers; $C(l_{\text{peak}})$ is the coherence at the peak layer; $\bar{C}$ is the global mean coherence across all layers; width is the region's layer count; and N is the model's layer count. The three factors have natural interpretations: the first normalizes prominence by the baseline signal level; the second (coherence boost) rewards peaks with above-average geometric organization, with a peak at the model-wide mean coherence scoring 2.0; the third rewards wider, more sustained regions while preventing broad shallow regions from dominating. The 0.5% floor was an initial empirical choice rather than a derived threshold; post-hoc ablation across detected gentle CAZes finds them causally active when suppressed — the validation paper [Henry, 2026c] confirms this at scale (full breakdown and threshold-dependence caveat in §7) — consistent with the floor admitting predominantly causal regions rather than noise. This is an upper-bound calibration: it confirms the floor is not too permissive, but does not establish whether it is too conservative — regions below the floor may also be causally active, and lower-bound calibration is open work. The scored detector is the primary detection method; the fixed-threshold detector is retained for backward compatibility.

Detection parameters. Three further parameters govern region detection and are held at their defaults throughout this paper: a minimum peak separation of 2 layers (`min_peak_distance`); a valley-merge threshold (`min_valley_depth_frac = 0.03`) that fuses adjacent regions whose separating valley is shallower than 3% of the global peak — on a merged region, prominence is recomputed as the peak height above the higher merged endpoint rather than the raw scipy saddle value, so the formula above applies to unmerged regions; and an optional architecture-calibrated re-ranking (`functional_caz_score`, which reweights the geometric score by attention-paradigm ablation priors) that exists in the library but is **not** applied to this paper's reported region counts. A reimplementer should set these to the stated defaults to reproduce the results here.

4.4 Multi-Layer Concept Extraction

The CAZ framework motivates the question: given that a concept assembles across multiple layers, where is the best layer to extract its probe direction? Three candidate methods are:

1. **Delta PCA**: PCA on layer-to-layer residual deltas $\Delta h_l = h_l - h_{l-1}$ within $[l_{\text{start}}, l_{\text{end}}]$. This captures what each layer *adds* — the construction process itself.
2. **Windowed PCA**: PCA on raw activations h_l across $[l_{\text{start}}, l_{\text{end}}]$. This captures the cumulative concept direction as it evolves through the allocation zone.
3. **Single-layer (baseline)**: The standard dom_vector at $l_{\max}$.

GEM continues this line of reasoning: the companion paper [Henry, 2026b], published simultaneously with this work, tracks the directional trajectory of a concept through the residual stream and shows that concept probe directions undergo substantial rotation *within* the CAZ, not settling into a stable encoding until a characteristic depth past the CAZ boundary — the *handoff layer*, derived from the same saddle-based segmentation defined in §4.3. Probing at the handoff layer is measurably more precise than peak-layer extraction in 341/493 concept × model trials (69.2%; model-level Wilcoxon W=356, N=29, one-sided). Excluding the two Gemma-2 models (whose single-direction ablation readout is known to fail on their distributed encoding rather than indicating a negative result, §7) raises this to 326/459 = 71.0%. A depth-matched control finds the settling contribution beyond

depth-of-extraction is weak (263/493 = 53.3% of pairs, not significant under any reported per-model aggregator): this is a depth effect, not a settling effect — probing deeper helps, independent of whether the deeper layer is where the direction specifically settles. The advantage holds regardless of attention architecture (MHA vs. GQA); an early architecture-dependent reading did not survive the expanded corpus ("Changes from Version 1"). The GEM paper has run the matched-layer delta-PCA comparison (near-equivalent to the difference-of-means probe, no layer effect established); windowed PCA remains distinct and unrun. Full results and the cohort breakdown are reported in [Henry, 2026b].

4.5 Sub-Representations

When a concept's $S(l)$ curve is multimodal, the dom_vector (the difference-in-means direction of §4.2 — the unit vector between class centroids, not a principal component) at each peak defines a distinct linear direction. Empirical measurement across 34 multimodal concept × model pairs (6 concepts — sentiment collapses to zero multimodal pairs at N=250; full breakdown in [Henry, 2026c] Table S6, §J) shows these directions are **geometrically distinct**: per-concept mean cosine between the shallow and deep peak dom_vectors falls in the range 0.12–0.41 — well above the random baseline ($|\cos| \approx 0.02$ for random unit vectors in these dimensions) but well below identity. The per-concept counts are small (under the N=250-sample scored detector, as few as 4 models for credibility), so the range endpoints should be interpreted cautiously. The two peaks are not the same direction at different amplitudes — they are candidate distinct linear features that both separate the same contrastive classes.

This implies that a single human concept label ("credibility", "negation") maps to **multiple sub-representations** at different processing depths. Interpretive evidence suggests:

- **Shallow sub-representations** form near the embedding layers, plausibly related to lexical cues (concept-associated words) — though a direct token-embedding alignment test does not confirm this (§5.6).
- **Deep sub-representations** form in mid-to-late layers, likely driven by compositional processing (contextual inference, scope, pragmatic reasoning).

Whether these are genuinely distinct sub-representations or a single direction rotating gradually across depth is not resolved by the present data. Layer × layer cosine similarity matrices show block-diagonal structure, and adjacent-layer cosine can dip at the saddle point (to as low as 0.35 in some models) — suggestive of a regime boundary — but an independent boundary check in the companion validation paper finds no reliable co-location between separation saddle points and direction-rotation events ([Henry, 2026c]). We therefore describe the two peaks as candidate distinct encodings without claiming an abrupt phase transition.

If sub-representations within a single model are geometrically distinct, the natural next question is whether they correspond *across* models: does a shallow sub-representation in one architecture align with a shallow sub-representation in another, and do deep sub-representations likewise pair? A monolithic single-rotation alignment between architectures would predict no such depth structure; a depth-stratified picture predicts that matched-depth pairs align more strongly than mismatched-depth pairs. This is the testable claim formalized as Prediction 5 (§5.5); a proper cross-architecture test requires alignment machinery beyond this paper's scope and is deferred to dedicated follow-up work.

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5. Testable Predictions

The CAZ framework generates seven predictions that are in principle falsifiable with existing open-weight models and standard interpretability tooling. Predictions were formulated from the theoretical framework prior to running any analysis; the broad phenomenon of multimodal allocation was observed early — preliminary profiles showed that some models produce multiple peaks — and this informed the framework's inclusion of scored detection; but the full quantitative scope (multimodal being the norm rather than the exception, mean 2.20 CAZes per concept per model) was not established until the validation pipeline ran. P4 in particular was formulated under the assumption that single-peak models were the primary case and multimodal profiles were exceptional. The specific directional predictions below were committed before per-concept results were examined.

Predictions P1–P7 were pre-specified on 2026-04-05 and published at https://waypoint.henrynet.ca/research/concept-assembly-zone/CAZ_Framework.pdf prior to running the full validation pipeline; they were not submitted to a formal external registry such as OSF. Verdicts were recorded after those analyses ran.

Verdicts use three categories: **Supported** (prediction holds), **Partially supported** (direction correct, stated mechanism or magnitude wrong), and **Not supported / Not testable as stated** (specific claim failed or premise invalidated). Predictions that cannot be given a clean verdict at this paper's scope are instead marked *Exploratory*, *Indeterminate*, or *Deferred* (defined in §8). Findings that emerged from investigating prediction failures — including results that held but were not pre-specified — are reported separately in §5.8 to keep the pre-specified record clean.

5.1 Optimal Ablation Depth

Prediction 1: The suppression-to-damage ratio varies systematically with intervention depth relative to the CAZ.

For a given concept, extract the concept direction at each layer via DoM. Apply orthogonal projection at each layer l independently. Measure: (a) behavioral suppression rate on targeted prompts, (b) KL divergence from the unmodified model on unrelated prompts. Plot the ratio (a)/(b) as a function of layer.

Status: Not supported. The prediction specified mid-CAZ as the optimal intervention point. Empirical results [Henry, 2026c] falsify this: in models with redundant encoding (Pythia, GPT-2, OPT), the concept direction persists throughout the post-allocation residual stream, making suppression layer-invariant; capability damage is lowest at late layers, so the optimal point is *post-CAZ*, not mid-CAZ. The specific claim failed. At its full 17-concept corpus scope the validation paper's own scorecard rates this prediction *partially supported* — the region-level claim holds while peak selection fails (§8). What investigating the failure revealed — a two-regime encoding structure that the prediction did not anticipate — is reported in §5.8. (Ablation coverage: §7.)

5.2 Cross-Architecture CAZ Ordering

Prediction 2: CAZ boundaries are concept-specific but show consistent relative ordering across architectures.

Different concepts should have different CAZ windows within the same model. However, the *relative* ordering of those windows — as a fraction of total model depth — should be consistent across

architectures. Absolute depth percentages may be family-specific, but relative concept ordering should be consistent.

Status: Not supported. A from-scratch recompute of this paper's own 7-concept ordering claim, against the 29 base models in the extraction corpus that carry all seven concepts, does not reproduce the originally published tendency. The recompute uses the current scored-detection methodology (0.5% prominence floor; depth read at each concept's *dominant CAZ region*, ranked by composite score per the finalized CAZ definition) — the same scored detector the validation paper uses, but not the same dominance convention: the validation paper's §3 ordering reads depth at the tallest-separation-peak region, while composite-score dominance is the convention of its §6.4. Median per-model $\tau = +0.05$ against the grand-mean consensus ordering (17 of 29 models positive, 12 negative; Wilcoxon signed-rank, two-sided $p = 0.54$ — not significant), with per-model τ spanning -0.71 to $+0.62$, roughly symmetric around zero rather than uniformly positive. This originally published figure (median $\tau = 0.473$, 26/26 models positive, $p = 1.49 \times 10^{-8}$) does not reproduce under the current pipeline; the exact methodology (detector variant, prominence floor, self-inclusion handling) that produced the original number is not fully recoverable from the published text alone, and the 7-concept ordering is in any case superseded by the validation paper's own from-scratch 17-concept analysis. That analysis ([Henry, 2026c] §3.1–3.2, 28 base models) finds a real but moderate tendency (median $\tau = 0.404$, 27 of 28 positive with gpt2-medium exactly zero and none negative, Wilcoxon $p = 1.49 \times 10^{-8}$) — read as *partially supported* there. The dominance convention accounts for part of the gap from that reading: on this paper's own C = 7 / 29-model data, reading depth at the tallest-separation-peak region — the convention the validation paper's §3 ordering uses — rather than at the composite-score dominant region raises the median to $\tau = +0.43$ (supported range). We report the composite-score result ($\tau = +0.05$, n.s.) as primary because composite-score dominance is the finalized CAZ convention, and disclose the metric sensitivity here rather than adopt the more favorable measure. At this paper's own 7-concept / 29-model scope the ordering prediction is **not supported**: it does not reproduce, and the point estimate is near zero (median $\tau \approx +0.05$, not significant). The validation paper's broader-corpus reading reaches *partially supported* (§8), but this paper's own data do not support the prediction. The consensus ordering (shallow → deep) spans a syntactic-to-behavioral gradient across 17 concepts from specificity to exfiltration (mean depths 21.4%–81.0%; [Henry, 2026c] §3). Detailed per-model breakdown in [Henry, 2026c].

5.3 CAZ Width and Concept Abstraction

Prediction 3: CAZ width correlates with concept abstraction level.

More abstract concepts (e.g., "trustworthiness," "moral valence") should have wider CAZ windows than concrete ones (e.g., "negation," "plurality"), because abstract concepts require more iterative construction across attention layers. This is testable by comparing $l_{\text{end}} - l_{\text{start}}$ for concepts at different levels of semantic abstraction as operationalized by, for example, depth in WordNet or scores on standard concreteness rating datasets.

Status: Exploratory — see §5.8. With $n = 6$ concepts and a researcher-assigned abstraction ranking, the sample is too small to treat this as a tested prediction; the result is reported in §5.8 as a hypothesis-generating finding. The validation paper's full 17-concept run reaches *partially supported* (§5.8, §8).

5.4 Post-CAZ Degradation as Logit Interference

Prediction 4: Post-CAZ re-entanglement correlates with unembedding matrix structure.

The degradation of clean concept geometry in late layers is not noise but a structural consequence of preparing the residual stream for logit projection. Concepts whose associated vocabulary tokens are distributionally similar in the unembedding space — close in embedding distance — should show more post-CAZ degradation than concepts with distributionally distinct vocabulary. This would explain why some concepts retain clean geometry into late layers (their vocabulary is well-separated) while others degrade early (their vocabulary clusters).

Status: Not testable as stated. P4 was formulated before the full quantitative scope of multimodal allocation was established — at that point, multi-peak profiles were seen in some models but not yet known to be the norm. The prediction assumes a single post-CAZ decay region, which is a unimodal premise. Multimodal allocation (mean 2.20 CAZes per concept per model; [Henry, 2026c]) invalidates that premise: under multiple allocation regions there is no single post-CAZ region against which to evaluate the unembedding correlation. (Whether the inter-peak decline reflects reallocation rather than degradation is a separate and unresolved question, §4.1; the verdict does not depend on it.) The prediction is not falsified by the data — its precondition is not met. A post-hoc analysis prompted by this failure is reported in §5.8.

5.5 Depth-Stratified Representational Convergence

Prediction 5: Cross-architecture alignment is depth-matched.

When a concept has multiple allocation regions, the sub-representation at a given processing depth should align more strongly with the corresponding-depth sub-representation in other architectures than with a different-depth sub-representation. Specifically, after Procrustes rotation, cosine(shallow_A, shallow_B) > cosine(shallow_A, deep_B) and cosine(deep_A, deep_B) > cosine(deep_A, shallow_B).

Status: Deferred — outside this paper's scope. The prediction concerns cross-architecture Platonic convergence: whether CAZ sub-representations align by processing depth across architectures. Evaluating it properly requires a dedicated cross-architecture alignment methodology with appropriate null controls. That machinery, and the prior question of whether any depth-matched signal is concept-specific rather than merely layer-specific, is the subject of dedicated follow-up work and is not adjudicated here: this framework paper neither confirms nor refutes P5. (v1 scored this prediction *Supported* on a peak-to-peak reading; this paper does not have the machinery to re-adjudicate that reading, so it withdraws the verdict and defers the question rather than restating it.)

Dataset note. The validation-paper-sourced C=17 figures this paper cites elsewhere — the P2 ordering $\tau = 0.404$ in §5.2 and the P3 width correlation $\rho = 0.307$ in §5.8 — are computed on the multi-model consensus Rosetta Concept Pairs dataset (14 generators — six model families across four AI labs; [Henry, 2026c]). This paper's *own* 7-concept computations (the τ null recompute in §5.2, the figures, and the worked examples in §6) use the single-model Claude Sonnet 4.6 concept dataset described in §7.

5.6 Lexical vs. Compositional Sub-Representations

Prediction 6: Shallow peaks encode lexical features; deep peaks encode compositional features.

The dom_vector at the shallow allocation peak should correlate with token embedding vectors for concept-associated words (e.g., "reliable", "dubious" for credibility). The dom_vector at the deep peak should show lower correlation with token embeddings and higher dependence on multi-token contextual patterns.

Status: Not supported. Token embedding probing — cosine similarity between peak dom_vectors and concept-relevant token embeddings — yields near-zero values (~ 0.02) at *both* the shallow and deep peaks. A paired two-sided Wilcoxon signed-rank test across the multimodal concept $\times$ model pairs (§4.5; $n = 34$) finds no difference between them ($p = 0.82$): the effect the prediction requires — shallow peaks more token-aligned than deep — is absent, and both peaks are near-orthogonal to raw token embeddings regardless. This is a small exploratory test at the 7-concept scale. The lexical/compositional distinction may operate at a higher level of abstraction than direct embedding alignment — the shallow feature could depend on token identity through multi-layer composition rather than literally pointing toward any single token's embedding vector. Testing this requires alternative experimental designs: per-token position attribution, attention knockout, or probing classifiers trained at each peak.

5.7 Multi-Modality as Architectural Property

Prediction 7: Multi-modality prevalence is determined by architecture, not scale.

The fraction of concepts showing multimodal $S(l)$ curves should vary more between architectural families (attention mechanism, activation function, training data) than between scales within a family.

Status: Indeterminate. The scale correlation is near zero ($\rho = 0.11$, $p = 0.63$, $n = 26$ models; the validation paper notes this correlation has not been recomputed on the full 28 [Henry, 2026c] §5.7), but at this n the test is underpowered — the 95% CI for ρ includes values up to ~0.47, so absence of a significant correlation is not evidence for the prediction. The qualitative family-level differences are real (Qwen 2.5: deep, prominent bimodality, valley depths 26–36%; Gemma 2: subtle structure below the 10% prominence threshold), but these are descriptive observations, not a formal test. Resolving P7 requires either a within-family scale ladder spanning a wider parameter range than the current Pythia ladder (8 sizes, 70M–12B — enough breadth in principle, but underpowered here for a formal architecture-vs-scale contrast) or a formally pre-registered architecture-level test against a stated alternative; both are open work [Henry, 2026c].

5.8 Findings Beyond the Predictions

Three results either emerged from investigating prediction failures or were not pre-specified to begin with. They are reported here separately to distinguish hypothesis-driven from data-driven findings.

Two-regime encoding structure (from investigating P1).

P1 predicted mid-CAZ ablation would be optimal; it was not (§5.1). Investigating why revealed a confound the prediction did not anticipate: encoding strategy determines whether layer choice matters at all. In models with redundant encoding (Pythia, GPT-2, OPT), the concept direction persists throughout the post-allocation residual stream, so suppression is layer-invariant. In models with sparse encoding (Qwen, Gemma), layer specificity is greater and CAZ location matters more. The two regimes are confounded with architecture vintage in this corpus (redundant-encoding models are older/absolute-position; sparse-encoding models are newer/RoPE-based), so encoding strategy specifically, as opposed to architecture family generally, is not isolated as the operative

variable. This is a productive falsification: the two-regime model is more explanatorily complete than the original prediction. Whether the CAZ profile *predicts* which regime applies before ablation — as opposed to describing it post hoc — is not established here: no held-out predictor was tested, and this is offered as a hypothesis for future validation [Henry, 2026c].

Chain-depth decay signal (post-hoc analysis prompted by P4).

P4's precondition — a single post-CAZ decay region — does not hold under multimodal allocation. After establishing this, we examined a redefined target: degradation from the *final* CAZ peak to the last layer. The validation paper's full-corpus test finds that remaining depth predicts decay ($r = -0.265$, $p = 4.08 \times 10^{-9}$; 476 measurements, 28 base models $\times$ 17 concepts; mean decay ratio 0.848) and, on the subset of 7 original concepts with hand-curated token lists (182 measurements, 26 models), unembedding token clustering shows a weak effect ($r = 0.173$, $p = 0.019$) [Henry, 2026c]. Neither set of measurements is independent — each shares models across concepts — so the pooled p-values are anticonservative; the effect sizes and their direction, not the p-values, are the evidentiary content. These results are consistent with the intuition behind P4 and are reported here as exploratory findings; they do not rescue P4 as stated.

Width-abstraction correlation (P3, exploratory).

Operationalization note: §5.3 proposed operationalizing "abstraction level" via depth in WordNet or scores on standard concreteness rating datasets. The executed test uses a researcher-assigned abstraction ranking instead. The proposed external validation (WordNet depth or concreteness ratings) was not carried out; the ranking was established by the author before examining per-concept widths.

Excluding credibility (bimodal, high variance), CAZ width correlates with researcher-assigned abstraction rank, with affective and epistemic concepts showing wider CAZes than relational and syntactic ones. This paper's original 6-concept test ($r = 0.294$, $n = 132$ concept-model pairs = 6 concepts $\times$ 22 models, permutation $p = 0.003$ on concept-level means) — nominally significant, but scored *exploratory* in the scorecard because $n = 6$ concepts is too small to constitute a test (§5.3, §8) — has since been superseded by the validation paper's full 17-concept run across 45 models (28 base + 9 instruct-tuned + 8 additional scale/frontier models — a wider corpus than the standard 28-base-model ablation set), which reaches the *partially supported* level: Spearman $\rho = 0.307$, $p = 3.7 \times 10^{-18}$ over $n = 765$ CAZ sweeps [Henry, 2026c]. Two caveats are carried from that validation-paper result: the abstraction ranking is author-assigned (external-ranking replication remains open), and the pooled p treats sweeps that share models and concepts as independent, so the modest effect size ($\rho = 0.307$) and its direction — not the pooled p — are the evidentiary content. The full 17-concept activation corpus (45 models) is available at the Rosetta Activations dataset [Henry, 2026, dataset].

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6. Proof of Concept

To demonstrate that the CAZ metrics and detection methods produce meaningful results, we present a minimal example on GPT-2-XL (48 layers, 1.5B parameters) using 7 concepts with 250 contrastive pairs each.

6.1 CAZ Detection

The separation curve $S(l)$ for credibility in GPT-2-XL peaks at layer 45 (94% depth) with $S = 1.17$ — the strongest signal among the 7 concepts tested (N=250 contrastive pairs per concept, $w = 3$ velocity smoothing, final-token probing). The scored detector identifies 2 CAZes for this concept under default detection settings (0.5% prominence floor, 3% valley-merge threshold; §4.3): a moderate early CAZ at L9 (19%, score=0.17) and the dominant peak at L45 (94%, major, score=0.84) — the dominant mode is the *deep* pole of credibility's bimodal profile (§4.5 multimodality), not a mid-network assembly event.

Across all 7 concepts in this single model, dominant allocation peaks span 27–94% depth (negation L13, temporal_order L21, certainty and moral_valence L23, sentiment L24, causation L25, credibility L45). Negation has two detected strong CAZes at L13 (27%, score=0.44) and L19 (40%, score=0.41) — close enough in score that which one is "dominant" is a near-tie, reflecting the shallow-to-deep sub-representation transition described in §4.5. Causation shows a single detected region at L25 under the current scored detector (a previously-observed shallow secondary region near L13 does not clear the detection threshold in the current corpus). The tendency for syntactic and relational concepts to peak earlier than epistemic ones holds for most of the set, though credibility is a clear exception in this model — consistent with P2's ordering tendency (§5.2) not replicating cleanly at this paper's original 7-concept scope, though the broader-corpus picture is more favorable (§5.2).

6.2 Scored Detection Reveals Hidden Structure

Lowering the detection threshold from 10% to 0.5% (scored detection) increases the number of detected CAZes from 7 to 10 in this single model — credibility, negation, and temporal_order each reveal secondary allocation regions invisible under the 10% threshold. The additional regions are not noise: the validation paper [Henry, 2026c] finds them causally active at scale (§7). The additional CAZes here correspond to interpretable sub-representation transitions (§4.5). GPT-2-XL is not the richest example of multimodal structure; models with more pronounced allocation dynamics (e.g., Pythia-160M, which shows a mean of 3.1 CAZes per concept across the 7 evaluated concepts) provide more vivid demonstrations — and the full cross-model pattern is reported in [Henry, 2026c]. Whether regions below the 0.5% floor are also causally active remains untested.

6.3 Scope of Validation

The framework has been evaluated across 35 models from 8 architectural families (Pythia, GPT-2, OPT, Qwen 2.5, Gemma 2, Llama 3.x, Mistral, Phi) spanning 70M to 14B parameters. Full empirical results — including multi-family scale ladders, structural analysis, and dark matter quantification — are reported in the validation paper [Henry, 2026c]. The reference implementation is provided as rosetta_tools v1.3.1 [Henry, 2026, software].

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7. Limitations

Known limitations include:

Synthetic contrastive data bias

This is the most significant limitation of the current work. All 7 concept datasets used for validation were generated by a single model (Claude Sonnet 4.6). Every CAZ location, every separation score,

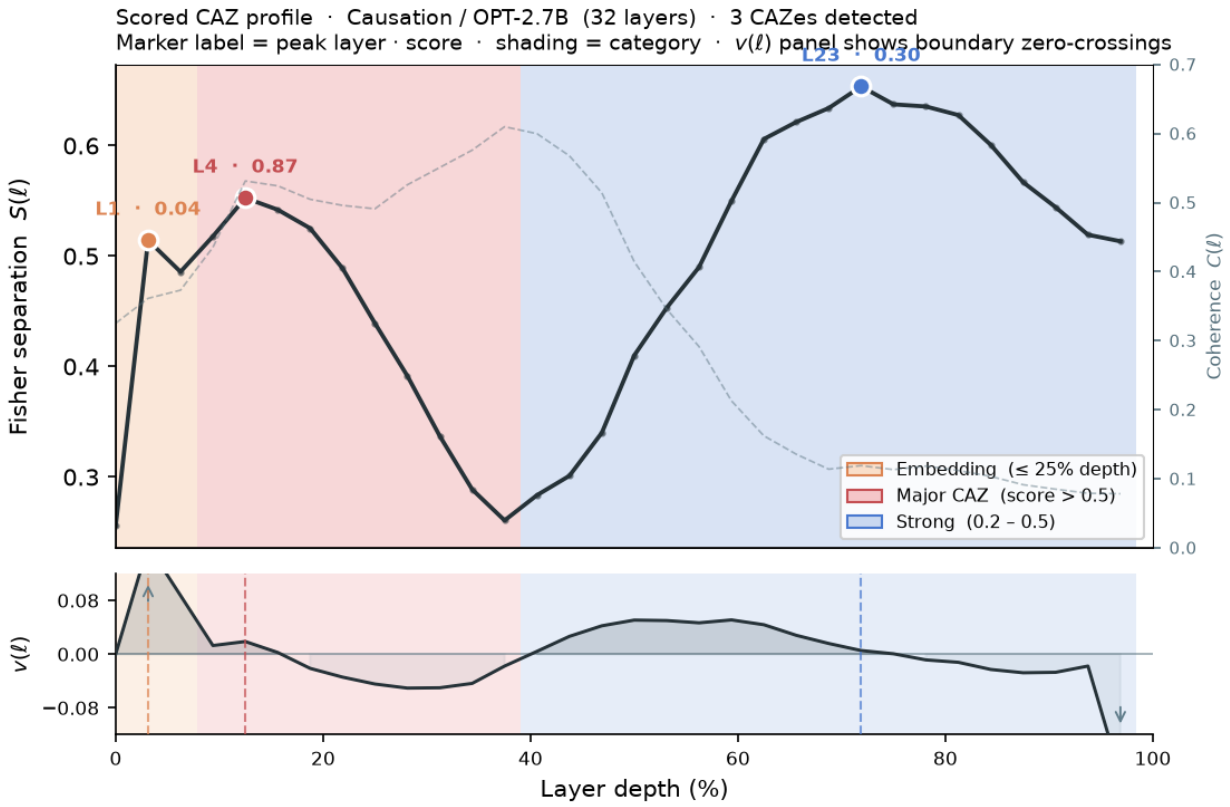

Figure 2: Scored CAZ profile for causation in OPT-2.7B (32 layers). Three CAZes detected: a gentle embedding CAZ at layer 1 (3% depth, score=0.04), a major CAZ at layer 4 (12% depth, score=0.87), and a strong CAZ at layer 23 (72% depth, score=0.30). The dashed line (right axis) is Coherence $C(l)$: it peaks near the saddle around 25% depth and is already declining by the deep separation peak at L23 — separation and coherence move independently (§4.2), and this profile is a direct illustration. N=250 contrastive pairs, $w = 3$ velocity smoothing (diagnostic only; boundaries are from the scored separation-curve detector), final-token probing, scored detection defaults. Depth percentages use the paper's l/N convention (§4.1), matching the rendered axis ticks.

and every depth estimate in this paper is measured against Claude's definition of each concept. If Claude's concept boundaries differ systematically from a human consensus definition, then what we are mapping is Claude's ontology projected onto other models, not a universal property of the models themselves. Multi-model consensus pair generation is addressed in the validation paper: Henry [2026c] §C replicates the peak-depth analysis on four architecturally diverse models using both the original single-generator pairs and the ~1,400-pair RCP consensus pool (14 generators — six model families across the Anthropic, Google, OpenAI, and Mistral labs), finding that the category-level ordering is preserved under both, while per-concept peak depths shift substantially in places (median $|\Delta| \approx 9.4$ pp). The framework methodology is independent of the data source; only the specific empirical results are at risk.

Pair fidelity

Human validation of the RCP pairs (the validation paper, Henry [2026c] §2.2/§8.8; a 31-rater crowd study, 540 analysis-grade ratings) confirms the corpus is high-fidelity for the large majority of concepts — 14 of 17 validate at $\geq$76%, though at the pair level only 66% of rated pairs are clean on this first pass — but not uniformly so. A minority, moral_valence most severely (no clean pair of ten), were built as opposite-pole (antonym) contrasts rather than presence-versus-absence contrasts, so their negatives still express the concept. Because CAZ extraction recovers whatever geometry the pairs make the model separate on, a bad pair still solidifies a stable, causally-active direction — just the wrong axis (for moral_valence, moral *polarity* rather than moral-valence *presence*). Per-concept results for these low-fidelity concepts should be read as geometry over the pole contrast, not concept presence; the aggregate results are carried by the high-fidelity majority and are unaffected. This is a property of the pair data, not the method — automated separability cannot catch it (a probe separates antonym pairs perfectly), and only human fidelity validation does. Henry [2026c] §8.8 develops the point.

Model scale

Full-corpus validation at N=250 extended coverage to the larger models — Pythia-12B, Gemma-2-9B, and Llama-3.1-8B — that were deferred from earlier reduced-scale runs. Frontier-scale models (70B+) were not evaluated in this paper's own CAZ runs; whether CAZ *structure* and the depth-ordering predictions hold at that scale is untested here.

Smoothing sensitivity

Under the scored detector used for all results here, CAZ boundaries are set by the separation curve's peak-prominence floor and valley-merge threshold (defaults 0.5% and 3%), not by the velocity smoothing width w or a velocity threshold: because the detector segments $S(l)$ directly, recomputing velocity at any smoothing window leaves the region counts and boundaries unchanged. The prominence and valley-merge defaults produce consistent results across 35 models but have not been formally validated against ground-truth concept boundaries.

Linearity assumption

The separation metric assumes the concept manifold is approximately linearly separable. For concepts with curved or multi-dimensional structure [Gurnee et al., 2026; Engels et al., 2025b], kernel-based or topological metrics may be required. This limitation is shared with most of the current interpretability literature.

Token position dependence

The framework probes a single fixed token position (the final token by default) and treats the

resulting layer-wise separation curve as representative of the concept. Zhao et al. [2025] demonstrate this directly for safety-relevant concepts: harmfulness encodes at the last instruction token while refusal encodes at the last post-instruction token — distinct positions for distinct concepts within the same prompt. CAZ boundaries derived from a single fixed token position may not generalize to concepts that encode elsewhere. No systematic sensitivity analysis on alternate token positions was conducted in this work; this is a known gap rather than an oversight, and position-varied replication is a concrete direction for follow-up.

Scope: semantic concepts only

The CAZ framework applies to concepts that are assembled through transformer layers via the allocation dynamics described here. Tokenization-level concepts — representations tied to byte-level patterns, script detection, BPE boundary artifacts, or sub-word orthography — do not undergo CAZ assembly in the same sense: they are largely established at the embedding layer and do not exhibit the velocity peaks, saddle-point transitions, or cross-architecture alignment that define the CAZ phenomenon. All seven concepts in the current validation corpus (credibility, certainty, causation, temporal_order, sentiment, negation, moral_valence) are semantic, and the framework's cross-model observations in this paper are scoped to that class. Extending CAZ analysis to tokenization-level representations requires a separate treatment of how such features behave under the layer-indexed metrics defined in §4.2.

Causal validation coverage

Ablation testing in this paper covers 16 of 35 models; the validation paper [Henry, 2026c] extends ablation coverage to all 28 base models (on the seven original concepts; the gentle-CAZ ablation table does not cover the ten later concepts) and confirms that 98% of gentle CAZes clear the validation paper's 20-percentage-point operating threshold, versus 28% of peak-distal layers — though the enrichment over peak-distal regions is threshold-dependent, ranging from ≈1.7× at a 5-percentage-point threshold to ≈20× at 50 ([Henry, 2026c] §8.7/§H), and the comparison is not fully independent of the selection criterion that defines CAZ vs. peak-distal layers in the first place (circularity caveat below). These causal claims are geometric (ablation-induced separation reduction); a single-layer behavioral pilot confirms the ablated directions are functionally active on next-token predictions (direction-specific in 27/28 models) but finds peak-vs-matched-control suppression indistinguishable from chance ([Henry, 2026c] §6.8) — so the framework's *peak-placement* causal support is geometric, not yet behavioral. The 6 instruct-tuned variants in this paper's corpus are not covered by [Henry, 2026c]'s base-model ablation sweep.

SAE residual cross-validation

§1 notes that CAZ regions may correspond to the nonlinear component of SAE reconstruction error — the residual that resists *linear* prediction from the input activation at a fixed layer, a minority of the error norm [Engels et al., 2025a]. The mechanism is plausible: in-progress concept construction within a CAZ produces transitional representations that are neither the input feature nor the output feature. This hypothesis has not been tested in this work. Direct cross-validation — measuring whether gentle CAZ layers predict elevated SAE unexplained residual in the same model — is a concrete open direction.

Construct and predictive limitations

Several results in this paper and its companions cut against the framework; they are consolidated here rather than left to the reader to assemble. (1) *Prediction scorecard*: of the seven pre-registered predictions, none is confirmed (full breakdown, §8). (2) *Score does not track causal*

importance: the CAZ score predicts geometric salience but not ablation impact — the validation paper documents higher-scored shallow CAZes with negligible suppression alongside lower-scored deep CAZes producing ~30% reduction ([Henry, 2026c] §6.1/§6.6), so the Major/Strong/Moderate/Gentle bands are a salience ordering, not a causal ranking. (3) *Gemma-2 readout failure*: Gemma-2 produces strong-range CAZ scores while single-direction ablation at the Fisher peak registers only weakly (mean 0.367, against 0.503 at CAZ peaks and 0.140 at peak-distal layers corpus-wide), and its cross-architecture ordering is not reliably estimated ([Henry, 2026c] §6.9). The validation paper shows this is what single-direction intervention *must* produce on a distributed code regardless of the underlying causal structure, so those cells are uninformative about Gemma-2 rather than evidence of causal inertness — its CAZes are unmeasured by this instrument, not shown non-causal (activation patching, which needs no direction estimate, recovers the concept near-completely there). (4) *Methodological reproducibility*: the v1 ordering statistic (§5.2) is not fully recoverable from the published method description alone. (5) *Null-model coverage*: the gentle-CAZ permutation null was evaluated on 5 of 28 base models, with the remainder assumed to generalize ([Henry, 2026c]). (6) *Circularity*: the construct is defined by the separation metric $S(l)$ (Fisher-style but not Fisher's criterion, §4.2), whose numerator is the class-centroid difference, and validated by projecting out that same direction and measuring the resulting drop in centroid separation, which annihilates the between-class mean at the ablated layer by construction. The non-tautological content is only what propagates to the *final* layer. Relatedly, peak-vs-peak-distal comparisons contrast layers selected for *high* separation against layers selected for *low* separation, so some positive gap is expected from the selection criterion alone; these causal claims should be read against that baseline, and the one non-circular test available (the behavioral peak-vs-matched-control pilot) is null ([Henry, 2026c] §6.8).

Pre-registration and instrument design

The seven predictions were pre-specified by the author on 2026-04-05 and published at `waypoint.henrynet.ca` prior to running the validation, but they were *not* deposited with an external registry (e.g., OSF), so the artifact carries no independent timestamp. More substantively, the measurement instrument is not fully independent of the test it feeds: the broad phenomenon of multimodal allocation was observed early in this corpus and informed the framework's adoption of scored detection (§5), so the detector that supplies the predictions' inputs was in part designed on the same data the predictions are evaluated against. The validation paper's from-scratch re-evaluation on the expanded corpus [Henry, 2026c] is the appropriate check against this (§8).

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8. Conclusion

The Concept Allocation Zone framework provides a methodology for tracking how concepts form across transformer depth — complementing the "best layer" view with a dynamical view of concept allocation.

The key contributions are:

1. **Three layer-wise metrics** (Separation, Coherence, Velocity) that characterize concept formation as a process, not a point.
2. **Scored detection** that reveals a spectrum of allocation regions from major to gentle, replacing binary thresholds with continuous scoring.
3. **The CAZ-is-not-a-concept distinction** — CAZes are depth-extended allocation segments

(residual-stream regions bounded by separation-curve saddles) in which the model organizes geometry to serve concepts. Multiple concepts share CAZes; single concepts participate in multiple CAZes across depth.

4. **Sub-representation tracking** across depth: within-model dom_vectors at shallow and deep peaks are not the same direction (cosine 0.12–0.41 with block-diagonal cosine similarity structure across layers, §4.5) — whether this reflects genuinely distinct sub-representations or one direction rotating gradually across depth is not resolved by the present data (§4.5). Whether these sub-representations correspond *across* architectures — the cross-architecture depth-matching question (P5) — is deferred to dedicated follow-up work (§5.5).

Verdict definitions. *Supported*: prediction holds. *Partially supported*: direction correct, stated mechanism or magnitude wrong. *Not supported*: specific claim failed. *Not testable as stated*: prediction's precondition invalidated by the data. *Exploratory*: result exists but sample too small to constitute a test. *Indeterminate*: the test as run is underpowered and does not resolve the prediction either way. *Deferred*: the prediction is outside this paper's evaluative scope and is assessed in dedicated follow-up work. Results arising from investigating prediction failures are in §5.8, not this table. The seven predictions are evaluated individually; no family-wise multiple-comparisons correction is applied at this paper's level (the validation paper applies one, [Henry, 2026c]).

#	Prediction	Verdict
P1	Mid-CAZ ablation yields optimal suppression-to-damage ratio	Not supported¶
P2	CAZ boundaries show consistent relative ordering cross-architecture	Not supported
P3	CAZ width correlates with abstraction level	Exploratory†
P4	Post-CAZ degradation correlates with unembedding matrix structure	Not testable as stated‡
P5	Cross-architecture alignment is depth-matched	Deferred§
P6	Shallow peaks encode lexical features; deep peaks encode compositional	Not supported
P7	Multi-modality prevalence is architectural, not scale-dependent	Indeterminate

¶See §5.8 (two-regime encoding structure). †See §5.8 (width-abstraction correlation). ‡See §5.8 (chain-depth decay post-hoc analysis). §Cross-architecture depth-matched convergence is outside this framework paper's scope and is evaluated in dedicated follow-up work; it is not scored here (§5.5). v1's *Supported* verdict rested on a peak-to-peak reading this paper does not re-adjudicate; the verdict is withdrawn and the question deferred.

Scope of these verdicts. These are the framework paper's per-prediction assessments at its original 7-concept evaluation scope. The validation paper [Henry, 2026c] re-evaluates all seven predictions on its full 17-concept, 28-base-model corpus (§5); where the two differ on scope — P1, P2, and P3 there reach *partially supported* — the validation paper's full-corpus evaluation supersedes the preliminary verdict here; on the remaining scored predictions the two papers agree (P4 not testable as stated, P6 not supported, P7 indeterminate). P5 is deferred rather than scored: the cross-architecture convergence question is outside this paper's scope and is the subject of dedicated follow-up work.

None of the seven pre-registered predictions is confirmed, and we present that as an honest result rather than a failure of the framework. The predictions were committed before the validation corpus existed; a framework whose purpose is to make concept formation measurable, and which states falsifiable predictions and reports that they do not survive contact with the data, has done its job. The framework's empirical footing is first-party and independent of the scorecard: the scored detector and its three metrics, the prevalence of multimodal allocation, the gentle-CAZ category, and — established in the validation paper [Henry, 2026c] — that the detected allocation structures are causally real in the geometric sense its protocol measures, read against the circularity, threshold-dependence, and behavioral-null caveats set out in §7. The most consequential revision the data required is the replacement of the single-peak assumption with multi-peak allocation: an *observed* finding that fed back into the detector design (§5), not a pre-registered prediction — it falsified prediction P4's single-peak premise and broadened the framework's descriptive scope. The same pattern runs through the scorecard: investigating prediction P1's failure surfaced the two-regime encoding structure (§5.8), and the scored detector's lower threshold surfaced the gentle-CAZ category — the falsifications are where the framework's richest current content came from. Full empirical results are reported in [Henry, 2026c].

The reference implementation is available as rosetta_tools v1.3.1 [Henry, 2026, software] (DOI 10.5281/zenodo.20361433), an open-source Python library providing the full CAZ extraction, alignment, ablation, and feature tracking pipeline described in this paper; the analysis, figure, and reproduction scripts are Rosetta_Analysis v1.1.0 (DOI 10.5281/zenodo.21583139), which imports that library, and the umbrella index for the series is at https://github.com/jamesrahenry/Rosetta. Pre-extracted model activations for all corpus models across 17 concepts are available as the Rosetta Activations dataset at https://huggingface.co/datasets/james-ra-henry/Rosetta-Activations (DOI 10.57967/hf/9725).

Changes from Version 1

This version corrects and retracts claims made in v1 (arXiv:2605.24856). A reader holding v1 should treat the following as superseded. Changes are grouped by *why* the value moved, because that distinction matters: some numbers changed because the corpus was rebuilt or grew, and others because v1 was wrong.

Verdict changes. Two of the seven pre-registered predictions changed verdict, and the change is consequential: v1 reported one prediction supported outright and one partially supported; **v2 confirms none** — one is deferred, the other is not supported at this paper's scope.

- **P5 (cross-architecture alignment is depth-matched): Supported → deferred.** v1 read 563/563 positive trials as confirming the pre-specified peak-to-peak claim. Properly evaluating cross-architecture Platonic convergence — including whether any alignment is anchored to CAZ *peaks*, as that reading assumed — needs dedicated cross-architecture alignment machinery and null controls that are outside this framework paper's scope. Rather than restate v1's verdict, P5 is **deferred to dedicated follow-up work** and is neither confirmed nor refuted here (§5.5).
- **P2 (consistent cross-architecture CAZ ordering): Partially supported → Not supported.** v1 reported median $\tau = 0.473$ (26/26 base models positive, $p = 1.49 \times 10^{-8}$). A from-scratch recompute under the current scored-detection pipeline, over the 29 base models carrying all seven concepts, does not reproduce it: median $\tau = +0.05$ (17/29 positive, p =

0.54). The exact v1 methodology is not fully recoverable from the published description. The validation paper's independent 17-concept analysis reaches *partially supported* and supersedes at that scope, but at this paper's own C = 7 scope the prediction is not supported (§5.2).

Claims withdrawn. Each was asserted in v1 and is no longer supported:

- **The instruction-tuning / alignment-training finding.** v1 (§1) asserted that instruction tuning "does not uniformly shift concept allocation to shallower depths; the direction and magnitude of change depend on the base model's existing concept geometry," and offered it as a CAZ-based metric of alignment-training distortion. The empirical claim is withdrawn to future work: the validation paper profiles base/instruct pairs (Supplementary §B) but no systematic base→instruct analysis was carried out. Only the in-principle metric remains, explicitly deferred (§1).
- **"Direct evidence of bandwidth reuse" (§4.1).** v1 called the four layer positions each hosting six of seventeen concept eigenvectors "direct evidence of bandwidth reuse." Downgraded: the validation paper flags this as a suggestive post-hoc observation lacking a permutation null, not an established convergence (§4.1).
- **The MHA/GQA handoff-cohort finding (§4.4).** v1 reported a pronounced, statistically significant architecture split on the 23-model GEM corpus — MHA models preferred handoff extraction at 85% versus 29% for GQA (Fisher's exact $p = 0.022$) — and stated "the CAZ profile predicts which case applies before extraction." On the expanded 29-model corpus this does not hold: MHA and GQA prefer handoff at statistically indistinguishable rates (76% vs. 67%, $p = 0.462$). The handoff-precision advantage itself is not withdrawn (it strengthens, 259/391 → 341/493 trials), but the architecture-dependence claim is: it was a small-corpus artifact, not a property of attention mechanism (§4.4).
- **Causation's bimodal split (§6.1).** v1's GPT-2-XL worked example reported causation splitting across a shallow region near L13 and its dominant peak at L25. Under the current scored detector and the true-N=250 corpus, causation shows a single detected region at L25; the shallow secondary region no longer clears the detection threshold. The list of concepts with secondary regions in §6.2 changes accordingly (causation → temporal_order).
- **The sub-representation "phase transition" (§4.5).** v1 described the shallow-to-deep transition at a CAZ's saddle point as "abrupt, not gradual… indicating a phase transition between distinct encoding regimes." An independent boundary check (validation paper) finds no reliable co-location between separation saddle points and direction-rotation events — indistinguishable from a permuted baseline. v2 withdraws the phase-transition characterization and describes the two peaks only as candidate distinct encodings.
- **"The CAZ profile predicts which regime applies" (§5.8).** The two-regime encoding structure (redundant- vs. sparse-encoding models, discovered investigating P1's failure) is a real, retained finding, but v1's implicit claim that the CAZ profile could predict which regime applies *before* ablation is run is not established: no held-out predictor was tested, and the two regimes are confounded with architecture vintage in this corpus.

Corrections — v1 was wrong or under-specified, independent of corpus size.

quantity	v1	v2	why
Mean CAZes per concept per model	3.4	**2.20**	v1 used the pre-N=250-rebuild corpus (mislabeled ≈ 200); recomputed on the true N = 250 corpus (§4.1)
Credibility worked example (§6.1)	peak **layer 29 (60%), S = 1.05**; dominant region score 0.37	**layer 45 (94%), S = 1.17**; dominant region score 0.84	true-N=250 rebuild; the dominant mode is now the deep pole of the bimodal profile
Negation worked example (§6.1)	peaks at L17 (35%, score 0.38) and L13 (28%, score 0.44)	peaks at L19 (40%, score 0.41) and L13 (27%, score 0.44) — near-tie for dominance	true-N=250 rebuild
Gentle-CAZ causal impact	"**93–100%** of cases"	**98%** at the 20-pp threshold (enrichment ≈1.7× at 5 pp → ≈20× at 50)	v1's band was the 16-model subset; recomputed on the validation paper's full 28-base-model sweep (§4.3)
Concept depth range (P2)	21.8%–84.7%	**21.4%–81.0%**	validation-paper recompute at C = 17
Velocity smoothing (captions, §6)	k = ⌊N/24⌋ (captions) and k = 2 (§6)	fixed default **w = 3**	the reported corpus uses the library default; CAZ boundaries come from the scored detector, not velocity (§4.2)
§4.5 sub-representation sample	46 concept×model pairs, 7 concepts, cosine range 0.156–0.433	34 pairs, 6 concepts (sentiment collapses to zero multimodal pairs at N=250), cosine range 0.12–0.41	true-N=250 rebuild
Abliteration capability effect (§3.3)	"independent replications… KL divergence 3.16–5.71… the intervention also affects general model capabilities"	Arditi et al. "report that this intervention disables refusal with minimal effect on other capabilities"	v1 mischaracterized the source's own reported result; corrected to what Arditi et al. actually report
SAE "dark matter" (§1, §2.1)	~50% of the SAE reconstruction error treated as unexplained "dark matter" CAZ may correspond to	Engels et al. find >90% of that error is *linearly predictable*; only the small residual *nonlinear* component is the candidate CAZ correspondence	v1 overstated the size of the phenomenon being explained

quantity	v1	v2	why
"Embedding CAZ" (Terminology, §4.1)	implied the literal embedding layer	denotes the first transformer block's *output*; the pre-transformer embedding is excluded from extraction	precision fix — affects how every Embedding CAZ reference should be read
Gurnee et al. [2026] citation (§2.2)	described the curved-manifold finding as **mid**-layer	corrected to **early**-layer, matching the source	citation-accuracy correction
Consensus-pair ordering check (§7)	"concept ordering is preserved across all four test architectures"	category-level ordering preserved, but **per-concept peak depths shift substantially in places** (median $\|\Delta\| \approx$ 9.4 pp)	validation paper's fuller RCP analysis added a caveat v1 didn't have

Figures — both changed subject model and were regenerated.

- **Figure 1 (right panel):** v1 showed *credibility in Qwen2.5-0.5B*; v2 shows *temporal_order in Pythia-1.4B*. "Black hole" terminology → "Major."
- **Figure 2:** v1 showed *negation in OPT-2.7B* (two "black-hole" CAZes plus a velocity panel); v2 shows *causation in OPT-2.7B* (three scored CAZes, no velocity panel).

Changes attributable to the corrected roster. The reported model count is **34** → **35** (the corpus is 29 base + 6 instruct; v1's "8 instruct" overcounted — only four families ship instruct variants). Validation-paper ablation coverage is **26** → **28** base models. The GEM handoff corpus is **23** → **29** base models (raising the headline handoff-precision trial count from 259/391 to 341/493). The P7 Pythia scale ladder is **7** → **8** sizes (70M–12B). The Rosetta Activations dataset now covers **33** → **46** language models (40 with full per-model analysis, 6 frontier models CAZ-only). The v1 hardware naming (L4/H200 GPUs, §7) was removed; GPU-hours were not logged (Disclosures).

New disclosures since v1 — not corrections, but not in v1 either. A reader relying on v1's causal picture should know v2 adds: (1) a **pair-fidelity audit** (§7) finding several concepts, moral_valence most severely, were built as antonym/opposite-pole contrasts rather than presence-vs-absence, so their geometry reflects polarity rather than the intended concept; (2) a **behavioral-null result** (§7): a single-layer pilot finds peak-vs-matched-control suppression indistinguishable from chance, so the framework's causal support is established only in the geometric sense, not yet behaviorally; (3) a consolidated **"Construct and predictive limitations"** subsection (§7) — CAZ score doesn't track causal importance, the Gemma-2 single-direction readout failure, a circularity concern in the $S(l)$ construct, and the gentle-CAZ null model having been checked on only 5 of 28 models; (4) a **"Pre-registration and instrument design"** self-critique (§7): the scored detector was in part designed on the same data the predictions are evaluated against. None of these change a v1 number — they're additions v1 did not have and could not have anticipated (most were added by the validation paper, [Henry, 2026c]) — but they materially change how strongly the causal claims in this paper should be read.

Full edit-level provenance is in `V2_CHANGELIST.md` in the accompanying repository.

Appendix: Common Definitions

This appendix collects the canonical cross-paper definitions shared by the Rosetta Program's Concept Allocation Zone (CAZ) framework (this paper), Geometric Evolution Maps (GEM) [Henry, 2026b], and CAZ Validation [Henry, 2026c]. It is a compact reference for a reader working from a single paper in isolation — each concept's full derivation, motivation, and supporting argument live in the paper cited alongside it, not here.

CAZ — Concept Allocation Zone. (Developed fully in §4 of this paper.)

> A CAZ is a locator: a coordinate on model depth that brackets where a concept is assembled — not the concept itself, and not a discrete functional module. Operationally, a CAZ is the segment of the residual stream between two saddle points of the separation curve $S(l)$, around a local separation peak. The detector partitions depth into such segments and scores each; the score grades geometric salience (Major→Gentle) — not causal importance, and not membership.

Three governing rules: (1) *No membership threshold* — every segment is a CAZ; "strong" vs. "gentle" is score, never a binary cut. (2) *Depth-extended, not depth-localized* — the segmentation tiles the full depth by construction; in the degenerate limit a single segment can span the entire network. (3) *A CAZ does not locate causation* — score is geometric salience, not causal importance; the causal work can be displaced from the separation peak and can sit outside the segment entirely.

GEM — Geometric Evolution Map. (Developed fully in the companion paper [Henry, 2026b] §3.)

> A GEM is the trajectory record of one concept's difference-of-means direction across one CAZ segment. The settled direction is its terminal reading, not a co-equal component.

Components: **window** — the layer span of the CAZ segment; **trajectory** — the per-layer dominant directions across that span, each estimated independently; **settled direction** — the trajectory's reading at the segment's final layer; **handoff layer** (L_H) — $\min(\text{end} + 1, N - 1)$, the first layer outside the segment, *derived* from the segment boundary and never independently detected by a rotation criterion.

Atlas. (Developed fully in [Henry, 2026b] §3.)

> A concept's atlas is the set of its GEMs in a model — one GEM per CAZ segment. An ordinary collection of maps: no transition structure between them is implied. An atlas is a container, not an operational object — it carries no defined rule for combining its GEMs into a single concept-level answer; any concept-level result applies its own aggregation rule, which must be stated wherever such a result is reported.

How they fit together. CAZ locates; GEM charts; the handoff is where a settled direction is first evaluated on activations it was not estimated on, and it sits outside the segment that produced it.

Vocabulary — use these, not the alternatives

term	means	do not say
CAZ / segment	a scored saddle-to-saddle segment; the locator	"the zone where the concept lives"; "depth-localized"
GEM	one segment's trajectory record	"the GEM" meaning a whole concept; "assembly event"
atlas	the set of a concept's GEMs in a model	"map" (collides with the M in GEM); "manifold"
settled direction	the trajectory's terminal reading, at segment's end	"the GEM probe"; "the direction at L_H"
handoff layer	segment.end + 1, derived	"detected where rotation ceases"; "the handoff within the zone"
caz_score	geometric salience, graded	"significance"; a membership threshold
peak-distal	the comparator layer/region away from a CAZ peak	"non-CAZ" — the tiling is total, so there is no non-CAZ layer in an absolute sense

Full provenance and the working argument behind these definitions: `DEFINITIONS.md` *and* `papers/shared/CAZ_DEFINITION_OF_RECORD.md` / `GEM_DEFINITION.md` *in the accompanying repository.*

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Self-evaluation. All predictions, experiments, analysis, and verdicts in this paper are the work of a single author; independent replication would carry more evidential weight (data and code are released to enable it). The double-exposure risk the validation paper flags applies here — several predictions share one source with their operationalization, and the scored detector was tuned in part on the evaluation corpus (§7, "Pre-registration and instrument design") — so the affirmed results should be read against §7's circularity caveat: the metric's non-tautological content is only what propagates to the final layer, and the one non-circular behavioral test available returned a null result.

Compute and licenses. GPU-hours and energy for extraction were not systematically logged. The corpus models carry a mix of licenses — several (Llama-3.x, Gemma-2) are released under non-OSI community licenses with use restrictions — and users should consult each model's license; the released artifacts (`rosetta_tools`, `Rosetta_Analysis`, and the activation dataset) are under their stated open licenses (see §8).

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References

- Alain, G., & Bengio, Y. (2017). Understanding intermediate layers using linear classifier probes. *arXiv preprint arXiv:1610.01644*. https://arxiv.org/abs/1610.01644
- Arditi, A., Obeso, O., Syed, A., Paleka, D., Panickssery, N., Gurnee, W., & Nanda, N. (2024). Refusal in language models is mediated by a single direction. *arXiv preprint arXiv:2406.11717*. https://arxiv.org/abs/2406.11717
- Belinkov, Y. (2022). Probing classifiers: Promises, shortcomings, and advances. *Computational Linguistics*, 48(1), 207–219.
- Belrose, N., Ostrovsky, I., McKinney, L., Furman, Z., Smith, L., Halawi, D., Biderman, S., & Steinhardt, J. (2023). Eliciting latent predictions from transformers with the tuned lens. *arXiv preprint arXiv:2303.08112*. https://arxiv.org/abs/2303.08112
- Bishop, C. M. (2006). *Pattern Recognition and Machine Learning*. Springer.
- Bricken, T., Templeton, A., Batson, J., Chen, B., Jermyn, A., Conerly, T., Turner, N., Anil, C., Denison, C., Askell, A., Lasenby, R., Wu, Y., Kravec, S., Schiefer, N., Maxwell, T., Joseph, N., Tamkin, A., Nguyen, K., McLean, B., Burke, J. E., Hume, T., Carter, S., Henighan, T., & Olah, C. (2023). Towards monosemanticity: Decomposing language models with dictionary learning. *Transformer Circuits Thread*, Anthropic. https://transformer-circuits.pub/2023/monosemantic-features/index.html
- Chan, L., Garriga-Alonso, A., Goldowsky-Dill, N., Greenblatt, R., Nitishinskaya, J., Radhakrishnan, A., Shlegeris, B., & Thomas, N. (2022). Causal scrubbing: A method for rigorously testing interpretability hypotheses. *AI Alignment Forum*, December 2022. https://www.alignmentforum.org/posts/JvZhhzycHu2Yd57RN/causal-scrubbing-a-method-for-rigorously-testing
- Cunningham, H., Ewart, A., Riggs, L., Huben, R., & Sharkey, L. (2023). Sparse autoencoders find highly interpretable features in language models. *arXiv preprint arXiv:2309.08600*. https://arxiv.org/abs/2309.08600
- Elhage, N., Nanda, N., Olsson, C., Henighan, T., Joseph, N., Mann, B., Askell, A., Bai, Y., Chen, A., Conerly, T., DasSarma, N., Drain, D., Ganguli, D., Hatfield-Dodds, Z., Hernandez, D., Jones, A., Kernion, J., Lovitt, L., Ndousse, K., Amodei, D., Brown, T., Clark, J., Kaplan, J., McCandlish, S., & Olah, C. (2021). A mathematical framework

for transformer circuits. *Transformer Circuits Thread*, Anthropic. https://transformer-circuits.pub/2021/framework/index.html

- Engels, J., Riggs, L., & Tegmark, M. (2025a). Decomposing the dark matter of sparse autoencoders. *Transactions on Machine Learning Research (TMLR)*, April 2025. *arXiv preprint arXiv:2410.14670*. https://arxiv.org/abs/2410.14670
- Engels, J., Michaud, E. J., Liao, I., Gurnee, W., & Tegmark, M. (2025b). Not all language model features are one-dimensionally linear. *Proceedings of the International Conference on Learning Representations (ICLR 2025)*. *arXiv preprint arXiv:2405.14860*. https://arxiv.org/abs/2405.14860
- Geva, M., Schuster, R., Berant, J., & Levy, O. (2021). Transformer feed-forward layers are key-value memories. *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP 2021)*, 5484–5495. *arXiv preprint arXiv:2012.14913*. https://arxiv.org/abs/2012.14913
- Geva, M., Caciularu, A., Wang, K. R., & Goldberg, Y. (2022). Transformer feed-forward layers build predictions by promoting concepts in the vocabulary space. *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP 2022)*, 30–45. *arXiv preprint arXiv:2203.14680*. https://arxiv.org/abs/2203.14680
- Goldowsky-Dill, N., MacLeod, C., Sato, L., & Arora, A. (2023). Localizing model behavior with path patching. *arXiv preprint arXiv:2304.05969*. https://arxiv.org/abs/2304.05969
- Gurnee, W., Horsley, T., Guo, Z. C., Kheirkhah, T. R., Sun, Q., Hathaway, W., Nanda, N., & Bertsimas, D. (2024). Universal neurons in GPT2 language models. *Transactions on Machine Learning Research (TMLR). arXiv preprint arXiv:2401.12181*. https://arxiv.org/abs/2401.12181
- Gurnee, W., Ameisen, E., Kauvar, I., Tarng, J., Pearce, A., Olah, C., & Batson, J. (2026). When models manipulate manifolds: The geometry of a counting task. *Transformer Circuits Thread*, Anthropic, October 2025 (arXiv archival 2026). *arXiv preprint arXiv:2601.04480*. https://arxiv.org/abs/2601.04480
- Henry, J. (2026, dataset). Rosetta Activations: Pre-extracted transformer residual stream activations for 46 language models across 17 concepts (40 with full per-model analysis; 6 frontier models CAZ-only). HuggingFace. https://huggingface.co/datasets/james-ra-henry/Rosetta-Activations (DOI: https://doi.org/10.57967/hf/9725)
- Henry, J. (2026, software). rosetta_tools (v1.3.1). Zenodo. https://doi.org/10.5281/zenodo.20361433
- Henry, J. (2026, software: Rosetta_Analysis). Rosetta_Analysis: Analysis, figure, and reproduction scripts for the Rosetta interpretability research program (v1.1.0). Zenodo. https://doi.org/10.5281/zenodo.21583139
- Henry, J. (2026b). Geometric Evolution Maps: Extracting Stable Concept Probes from Transformer Residual Streams. *arXiv preprint arXiv:2605.25848*. https://arxiv.org/abs/2605.25848
- Henry, J. (2026c). Concept Encoding Strategies Across 28 Transformers: A Concept Allocation Zone and Geometric Evolution Map Evaluation. Zenodo. https://doi.org/10.5281/zenodo.22016917
- Huh, M., Cheung, B., Wang, T., & Isola, P. (2024). Position: The Platonic Representation Hypothesis. *Proceedings of the 41st International Conference on Machine Learning (ICML 2024)*,

PMLR 235, 20617–20642. *arXiv preprint arXiv:2405.07987*. https://arxiv.org/abs/2405.07987

- Kim, B., Wattenberg, M., Gilmer, J., Cai, C., Wexler, J., Viégas, F., & Sayres, R. (2018). Interpretability beyond feature attribution: Quantitative testing with concept activation vectors (TCAV). In *Proceedings of the 35th International Conference on Machine Learning (ICML 2018)*, 2668–2677. *arXiv preprint arXiv:1711.11279*. https://arxiv.org/abs/1711.11279
- Kornblith, S., Norouzi, M., Lee, H., & Hinton, G. (2019). Similarity of neural network representations revisited. *Proceedings of the International Conference on Machine Learning (ICML 2019)*, 3519–3529.
- Mahalanobis, P. C. (1936). On the generalized distance in statistics. *Proceedings of the National Institute of Sciences of India*, 2(1), 49–55.
- Marks, S., & Tegmark, M. (2024). The geometry of truth: Emergent linear structure in large language model representations of true/false datasets. *Conference on Language Modeling (COLM 2024)*. *arXiv preprint arXiv:2310.06824*. https://arxiv.org/abs/2310.06824
- Meng, K., Bau, D., Andonian, A., & Belinkov, Y. (2022). Locating and editing factual associations in GPT. *Advances in Neural Information Processing Systems (NeurIPS 2022)*, 35, 17359–17372. *arXiv preprint arXiv:2202.05262*. https://arxiv.org/abs/2202.05262
- nostalgebraist. (2020). Interpreting GPT: The logit lens. *LessWrong / AI Alignment Forum*, August 2020. https://www.lesswrong.com/posts/AcKRB8wDpdaN6v6ru/interpreting-gpt-the-logit-lens
- Sofroniew, N., Kauvar, I., Saunders, W., Chen, R., Henighan, T., Hydrie, S., Citro, C., Pearce, A., Tarng, J., Gurnee, W., Batson, J., Zimmerman, S., Rivoire, K., Fish, K., Olah, C., & Lindsey, J. (2026). Emotion concepts and their function in a large language model. *Transformer Circuits Thread*, Anthropic. *arXiv preprint arXiv:2604.07729*. https://arxiv.org/abs/2604.07729
- Tenney, I., Das, D., & Pavlick, E. (2019). BERT rediscovers the classical NLP pipeline. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL 2019)*, 4593–4601. *arXiv preprint arXiv:1905.05950*. https://arxiv.org/abs/1905.05950
- Vig, J., Gehrmann, S., Belinkov, Y., Qian, S., Nevo, D., Singer, Y., & Shieber, S. (2020). Investigating gender bias in language models using causal mediation analysis. *Advances in Neural Information Processing Systems (NeurIPS 2020)*, 33, 12388–12401.
- Wang, K., Variengien, A., Conmy, A., Shlegeris, B., & Steinhardt, J. (2023). Interpretability in the wild: A circuit for indirect object identification in GPT-2 small. *Proceedings of the International Conference on Learning Representations (ICLR 2023)*. *arXiv preprint arXiv:2211.00593*. https://arxiv.org/abs/2211.00593
- Wollschläger, T., Elstner, J., Geisler, S., Cohen-Addad, V., Günnemann, S., & Gasteiger, J. (2025). The geometry of refusal in large language models: Concept cones and representational independence. *Proceedings of Machine Learning Research (ICML 2025)*, 267, 66945–66970. *arXiv preprint arXiv:2502.17420*. https://arxiv.org/abs/2502.17420
- Zhao, J., Huang, J., Wu, Z., Bau, D., & Shi, W. (2025). LLMs encode harmfulness and refusal separately. *arXiv preprint arXiv:2507.11878*. https://arxiv.org/abs/2507.11878
- Zou, A., Phan, L., Chen, S., Campbell, J., Guo, P., Ren, R., Pan, A., Yin, X., Mazeika, M., Dombrowski, A.-K., Goel, S., Li, N., Byun, M. J., Wang, Z., Mallen, A., Basart, S.,

Koyejo, S., Song, D., Fredrikson, M., Kolter, J. Z., & Hendrycks, D. (2023). Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*. https://arxiv.org/abs/2310.01405